POWER ELECTRONICS

Module 3: AC Voltage Controllers, DC-AC Inverters & Utility Interface

B.Tech S5 Electrical & Electronics Engineering (EEE) · APJ Abdul Kalam Technological University

Comprehensive Master Study Guide & University Examination Preparation Manual

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AC Voltage Controllers (ACVC) / AC Regulators

Foundational Principles and Classification

An **AC Voltage Controller** (also termed an **AC Voltage Regulator**) is a line-frequency phase-controlled or integral-cycle switched converter that converts a fixed-voltage, fixed-frequency alternating current supply into a **variable-voltage, fixed-frequency AC output** without an intermediate DC stage:

$$\text{Fixed AC Input } (V_s, f) \xrightarrow{\text{AC Voltage Controller}} \text{Variable AC Output } (V_{or}, f) \tag{1}$$

Because conversion is accomplished entirely in a single stage using line-commutated thyristors or TRIACs, AC voltage controllers feature extremely high operational efficiency ($\eta > 98\%$), rugged mechanical simplicity, low initial cost, and the complete absence of bulky reactive filter banks.

Classification of AC Voltage Control Methods

AC voltage controllers are categorized into three distinct operational methodologies according to the switching mechanism employed (frequently evaluated in university examinations):

1. **Phase Angle Control (PAC):**
   - **Mechanism:** The power semiconductor switches (a pair of anti-parallel SCRs or a TRIAC) connect the load to the AC source for a controllable fraction of each supply half-cycle. Conduction is initiated at a controllable delay angle α (measured relative to the zero-crossing of the input supply voltage) and ceases naturally at current zero (β) via natural (line) commutation.
   - **Advantages:** Provides continuous, smooth, instantaneous, and infinitely variable control of RMS load voltage from 0 to V_s.
   - **Disadvantages:** The abrupt chopping of sinusoidal voltage waveforms introduces substantial low-order current harmonics (3rd, 5th, 7th) and causes the supply displacement factor to lag significantly as α increases.
   - **Primary Applications:** Universal motor speed control, light dimmers, induction motor soft-starters, industrial furnace heating.
2. **Integral Cycle Control (On-Off / Burst Firing Control):**
   - **Mechanism:** The thyristor switches connect the load to the AC mains for an integral number of complete supply cycles (n cycles), and then remain un-triggered for another integral number of cycles (m cycles). Switching is always synchronized to occur at the zero-crossing of the supply voltage ($v_s(t) = 0$).
   - **Mathematical Duty Ratio and RMS Voltage:**

$$k = \frac{n}{n+m} = \frac{T_{on}}{T_o} \implies V_{or} = V_s\sqrt{k} = V_s\sqrt{\frac{n}{n+m}} \tag{2}$$

     where $T_o = (n+m)T$ is the overall repeating period.
   - **Advantages:** Because switching transitions occur exclusively at natural voltage zero-crossings ($dv/dt \to 0$ and $di/dt \to 0$), radio frequency interference (RFI), electromagnetic interference (EMI), and switching losses are virtually zero.
   - **Disadvantages:** Sub-harmonic frequency components (frequencies below the mains supply frequency f) and line flicker are introduced, making this method completely unsuitable for motor speed control or lighting systems.
   - **Primary Applications:** Loads with very large mechanical or thermal time constants, such as industrial electric ovens, boiler heaters, and plastic extrusion mold heaters.
3. **Sequence Control of AC Voltage Controllers:**

- **Mechanism:** Two or more parallel transformer-tapped stages of AC voltage controllers are fired in sequence. By controlling only one stage at a time while leaving others fully ON or fully OFF, the output voltage is varied with substantially reduced harmonic distortion and a significantly improved supply power factor across the entire operating range.

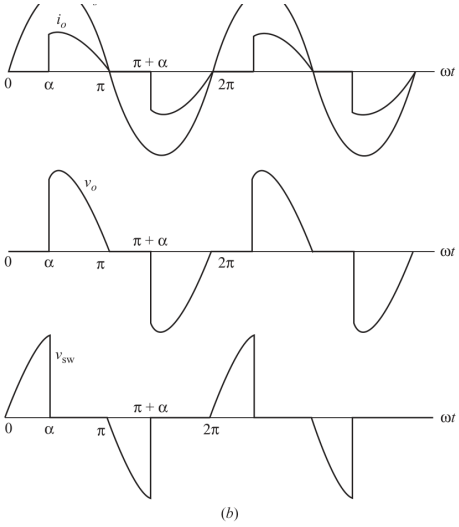


Figure 5-1 (*a*) Single-phase ac voltage controller with a resistive load; (*b*) Waveforms.

Figure 1: Single-phase full-wave AC voltage controller with resistive load: (a) Power circuit diagram using two anti-parallel thyristors T_1 and T_2; (b) Supply voltage v_s, gating signals i_{g1} and i_{g2}, load voltage v_o, and thyristor voltage v_{T1} waveforms for firing delay angle α. (Source: Daniel W. Hart, Fig. 5-1)

Single-Phase Full-Wave AC Voltage Controller with Pure Resistive (R) Load

Consider the circuit of Figure 1a supplied by a single-phase sinusoidal AC source $v_s(t) = V_m \sin(\omega t) = \sqrt{2}V_s \sin(\omega t)$. Two SCRs ($T_1$ and T_2) are connected in anti-parallel (back-to-back) between the AC mains and a resistive load R.

Circuit Operation Across One Full Fundamental Cycle

- **Positive Half-Cycle ($0 \le \omega t \le \pi$):** Supply voltage $v_s > 0$. Thyristor T_1 is forward-biased, while T_2 is reverse-biased ($v_{AK2} < 0$).
  - From $\omega t = 0$ to $\omega t = \alpha$, T_1 remains OFF. No current flows ($i_o = 0$), load voltage $v_o = 0$, and the full line voltage appears across T_1 ($v_{T1} = v_s$).
  - At $\omega t = \alpha$, a firing pulse is applied to the gate of T_1. T_1 turns ON instantaneously.
  - From $\omega t = \alpha$ to π, T_1 conducts load current:

$$v_o(t) = v_s(t) = V_m \sin(\omega t), \quad i_o(t) = \frac{v_s(t)}{R} = \frac{V_m}{R} \sin(\omega t) \tag{3}$$

  - At $\omega t = \pi$, the supply voltage reverses polarity. The current through T_1 drops naturally to zero ($i_o(\pi) = 0$), turning T_1 OFF via natural (line) commutation.
- **Negative Half-Cycle ($\pi \le \omega t \le 2\pi$):** Supply voltage $v_s < 0$. Thyristor T_2 is forward-biased, while T_1 is reverse-biased.
  - At $\omega t = \pi + \alpha$, a firing pulse is applied to the gate of T_2. T_2 turns ON.

- From $\omega t = \pi + \alpha$ to 2π, T_2 conducts negative load current:

$$v_o(t) = v_s(t) = V_m \sin(\omega t), \quad i_o(t) = \frac{V_m}{R}\sin(\omega t) \quad (< 0) \tag{4}$$

- At $\omega t = 2\pi$, $i_o(2\pi) = 0$, and T_2 turns OFF naturally.

Rigorous Mathematical Derivation of RMS Output Voltage

Because the output voltage waveform exhibits half-wave symmetry ($v_o(\omega t + \pi) = -v_o(\omega t)$), the root-mean-square (RMS) value can be computed over a half-cycle (0 to π):

$$\begin{aligned} V_{or} &= \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi} [V_m \sin(\omega t)]^2 \, d(\omega t)} = \sqrt{\frac{V_m^2}{\pi}\int_{\alpha}^{\pi} \frac{1-\cos(2\omega t)}{2} d(\omega t)} \\ &= \sqrt{\frac{V_m^2}{2\pi}\left[\omega t - \frac{\sin(2\omega t)}{2}\right]_{\alpha}^{\pi}} = \sqrt{\frac{V_m^2}{2\pi}\left[(\pi-\alpha) - \left(\frac{\sin(2\pi)-\sin(2\alpha)}{2}\right)\right]} \\ &= \sqrt{\frac{V_m^2}{2\pi}\left[(\pi-\alpha) + \frac{\sin(2\alpha)}{2}\right]} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{1}{\pi}\left[(\pi-\alpha) + \frac{\sin(2\alpha)}{2}\right]} \end{aligned} \tag{5}$$

Recognizing that $V_s = V_m/\sqrt{2}$ is the RMS value of the input source:

Master Equation: RMS Output Voltage (R Load)

$$V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{6}$$

Performance Characteristics with R Load

- **RMS Load Current:** $I_{or} = \frac{V_{or}}{R} = \frac{V_s}{R}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}$.
- **Average and RMS Thyristor Currents:** Each thyristor conducts for only one half of the symmetrical period:

$$I_{T(avg)} = \frac{1}{2\pi}\int_{\alpha}^{\pi}\frac{V_m}{R}\sin(\omega t)d(\omega t) = \frac{V_m}{2\pi R}(1+\cos\alpha) = \frac{\sqrt{2}V_s}{2\pi R}(1+\cos\alpha) \tag{7}$$

$$I_{T(rms)} = \frac{I_{or}}{\sqrt{2}} = \frac{V_s}{\sqrt{2}R}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{8}$$

- **Input Power Factor (PF):** Since input current $i_s(t) = i_o(t)$, the apparent power supplied by the mains is $S = V_s I_{or}$. The active power dissipated in the load is $P_o = I_{or}^2 R = \frac{V_{or}^2}{R}$.

$$\text{PF} = \frac{P_o}{S} = \frac{V_{or}^2/R}{V_s I_{or}} = \frac{V_{or} I_{or}}{V_s I_{or}} = \frac{V_{or}}{V_s} = \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{9}$$

- **Control Range:** As α varies from 0 to π ($180°$):
  - At $\alpha = 0$: $V_{or} = V_s$, PF $= 1.0$ (Full sinusoidal conduction).
  - At $\alpha = \pi/2$ ($90°$): $V_{or} = V_s/\sqrt{2} \approx 0.707 V_s$, PF ≈ 0.707.
  - At $\alpha = \pi$ ($180°$): $V_{or} = 0$, PF $= 0$ (Output completely blocked).

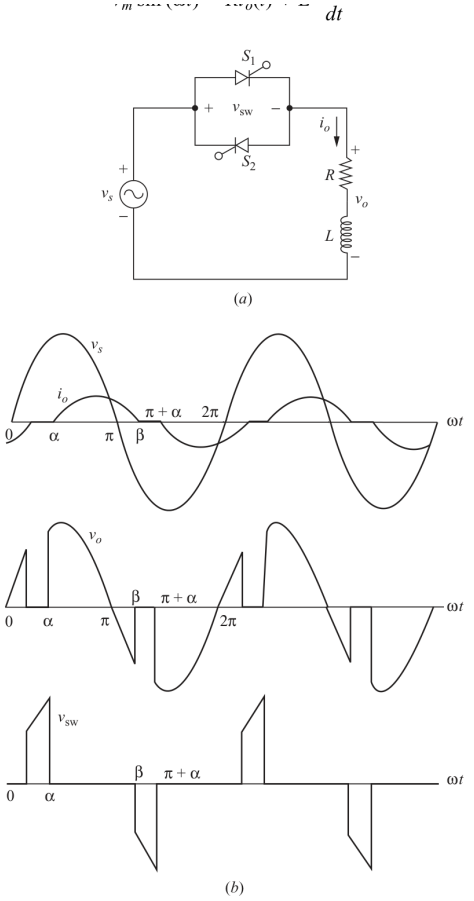


Figure 5-4 (*a*) Single-phase ac voltage controller with an *RL* load; (*b*) Typical waveforms.

Figure 2: Single-phase AC voltage controller with RL load: Waveforms of input voltage v_s, thyristor gating pulses, output voltage v_o, and inductive load current $i_o(\omega t)$ showing the extinction angle β and conduction angle $\gamma = \beta - \alpha$. (Source: Daniel W. Hart, Fig. 5-4)

Single-Phase Full-Wave AC Voltage Controller with RL Load

When the load consists of an inductive impedance $Z = \sqrt{R^2 + (\omega L)^2}$ with characteristic load impedance angle $\phi = \arctan\left(\frac{\omega L}{R}\right)$, the inductance opposes sudden changes in load current. Consequently, **the current continues to flow after the source voltage reverses polarity at** $\omega t = \pi$.

Derivation of the Load Current Differential Equation

During the conduction of thyristor T_1 ($\alpha \leq \omega t \leq \beta$), the governing Kirchhoff's Voltage Law (KVL) differential equation is:

$$L\frac{di_o(t)}{dt} + Ri_o(t) = v_s(t) = V_m \sin(\omega t) \tag{10}$$

The total solution consists of a steady-state particular solution $i_{ss}(t)$ and a transient complementary solution $i_{tr}(t)$:

$$i_o(\omega t) = \frac{V_m}{Z}\sin(\omega t - \phi) + Ae^{-\frac{R}{\omega L}\omega t} \tag{11}$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ and $\phi = \arctan(\omega L/R)$.

Applying the initial boundary condition at the firing instant: at $\omega t = \alpha$, $i_o(\alpha) = 0$:

$$0 = \frac{V_m}{Z}\sin(\alpha - \phi) + Ae^{-\frac{R}{\omega L}\alpha} \implies A = -\frac{V_m}{Z}\sin(\alpha - \phi)e^{\frac{R}{\omega L}\alpha} \tag{12}$$

Substituting A back into the current expression yields the closed-form equation for the instantaneous load current:

Instantaneous Load Current in ACVC with RL Load

$$i_o(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \phi) - \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\omega t - \alpha)}\right], \quad \text{for } \alpha \leq \omega t \leq \beta \tag{13}$$

Extinction Angle (β) and Conduction Angle (γ)

At $\omega t = \beta$, the magnetic energy stored in inductor L is completely dissipated into R, and the current extinguishes to zero ($i_o(\beta) = 0$). Setting equation (13) to zero at $\omega t = \beta$ gives the transcendental equation:

$$\sin(\beta - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\beta - \alpha)} \tag{14}$$

The conduction angle of each thyristor is defined as $\gamma = \beta - \alpha$. Equation (14) can be expressed in terms of γ as:

$$\sin(\alpha + \gamma - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}\gamma} \tag{15}$$

Three Critical Conduction Regimes Based on Firing Angle α vs ϕ

The behavior of the RL AC voltage controller depends decisively on the relative relationship between the firing delay angle α and the load impedance phase angle ϕ:

1. **Case 1: $\alpha > \phi$ (Discontinuous Conduction Mode — Standard Operation):**
   - The current extinguishes before the next half-cycle thyristor is triggered: $\beta < \pi + \alpha$, which means the conduction angle $\gamma = (\beta - \alpha) < \pi$.
   - There is a finite zero-current interval between the extinction of T_1 and the firing of T_2.
   - The output voltage is fully controllable by varying α between ϕ and π.
2. **Case 2: $\alpha = \phi$ (Boundary / Critical Conduction Mode):**

- Substituting $\alpha = \phi$ into equation (14):

$$\sin(\beta - \phi) = \sin(0) \cdot e^{-\frac{R}{\omega L}(\beta-\phi)} = 0 \implies \beta - \phi = \pi \implies \beta = \pi + \phi = \pi + \alpha \tag{16}$$

- The conduction angle is exactly $\gamma = \beta - \alpha = \pi$ (180°).
- Thyristor T_1 conducts for exactly 180°. The moment T_1 extinguishes at $\omega t = \pi + \alpha$, thyristor T_2 is triggered!
- The load current becomes a continuous, perfectly sinusoidal waveform: $i_o(t) = \frac{V_m}{Z}\sin(\omega t - \phi)$. The converter behaves as if the load is directly connected across the AC supply!

3. **Case 3: $\alpha < \phi$ (Dangerous / Loss of Control Mode):**

- If a narrow gate pulse is applied at $\alpha < \phi$, thyristor T_1 cannot turn ON if the previous thyristor T_2 is still conducting negative current!
- If continuous or long gate pulses are used, T_1 will not turn ON until T_2 current reaches zero at β. This effectively clamps the minimum firing angle to $\alpha_{\text{min}} = \phi$.
- **Critical Exam Rule:** Firing angle α must always satisfy $\alpha \geq \phi$. A firing angle smaller than ϕ does not increase the load voltage and causes a loss of independent phase control or asymmetrical DC saturation of feeding transformers!

Derivation of RMS Output Voltage with RL Load

During the conduction interval $\alpha \leq \omega t \leq \beta$, the load voltage equals the supply voltage: $v_o(t) = V_m \sin(\omega t)$. Outside this interval, $v_o(t) = 0$.

$$\begin{aligned} V_{or} &= \sqrt{\frac{1}{\pi}\int_{\alpha}^{\beta} [V_m \sin(\omega t)]^2 \, d(\omega t)} = \sqrt{\frac{V_m^2}{2\pi}\int_{\alpha}^{\beta} [1 - \cos(2\omega t)] \, d(\omega t)} \\ &= \sqrt{\frac{V_m^2}{2\pi}\left[(\beta - \alpha) - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} \end{aligned} \tag{17}$$

In terms of $V_s = V_m/\sqrt{2}$ and conduction angle $\gamma = \beta - \alpha$:

Master Equation: RMS Output Voltage (RL Load)

$$V_{or} = V_s\sqrt{\frac{1}{\pi}\left[(\beta - \alpha) - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} = V_s\sqrt{\frac{1}{\pi}\left[\gamma - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]} \tag{18}$$

Single-Phase AC Voltage Controller with Pure Inductive (L) Load

For a purely inductive load ($R = 0 \implies \phi = 90° = \pi/2$):

$$L\frac{di_o}{dt} = V_m \sin(\omega t) \implies i_o(\omega t) = \frac{V_m}{\omega L}\int_{\alpha}^{\omega t} \sin(\omega t) d(\omega t) = \frac{V_m}{\omega L}(\cos\alpha - \cos\omega t) \tag{19}$$

The current extinguishes when $i_o(\beta) = 0$:

$$\cos\alpha - \cos\beta = 0 \implies \beta = 2\pi - \alpha \tag{20}$$

The conduction angle is:

$$\gamma = \beta - \alpha = 2\pi - 2\alpha = 2(\pi - \alpha) \tag{21}$$

- Since $\phi = \pi/2$ (90°), the control range for α is restricted to $\pi/2 \leq \alpha \leq \pi$.

- At $\alpha = \pi/2$: $\beta = 2\pi - \pi/2 = 3\pi/2, \gamma = \pi$ (180°). Current is continuous.
- At $\alpha = \pi$: $\beta = \pi, \gamma = 0$. No current flows.
- This configuration is the fundamental building block of **Thyristor-Controlled Reactors (TCR)** utilized extensively in static VAR compensators (SVC) on power transmission grids!

Gate Firing Pulse Requirements for Inductive Loads

With an inductive load, a standard short gate trigger pulse ($10 - 20\ \mu$s) cannot ensure reliable thyristor turn-on:

Gate Firing Requirements with Inductive Loads

Because $di/dt = v_s/L$ is restricted by the load inductance, the anode current rises slowly after triggering. If a short gate pulse ends before the anode current crosses the **latching current** (I_L), the thyristor will immediately turn back OFF when the gate pulse is removed!
Furthermore, due to current overlap between consecutive half-cycles, the incoming thyristor may be triggered while the outgoing thyristor is still conducting.
Mandatory Solutions:

1. **Continuous Gate Gating:** Maintain gate current throughout the desired conduction period (α to π). (Drawback: High average gate power dissipation).
2. **High-Frequency Pulse Train Gating (Carrier Frequency Modulation):** Modulate the firing signal with a $10 - 30$ kHz pulse train generated by a 555 timer or microcontroller. This reduces gate power dissipation by $> 80\%$ while allowing small, compact pulse isolation transformers!

Engineering Applications of AC Voltage Controllers

1. **Industrial and Domestic Heating:** Closed-loop temperature regulation in resistance furnaces, kilns, soldering stations, and water heaters.
2. **Lighting Control (Dimmers):** Theater stage lighting, architectural illumination, and incandescent lamp dimming.
3. **AC Motor Soft Starters:** During the starting of high-power three-phase squirrel-cage induction motors, direct-on-line (DOL) starting draws $6 - 8 \times I_{\text{rated}}$, creating severe line voltage dips. By ramping α from $150°$ down to ϕ, starting current is limited to $2 - 3 \times I_{\text{rated}}$.
4. **Speed Control of AC Fan and Pump Motors:** Controlling stator voltage on high-slip single-phase and three-phase induction motors provides stable variable-speed torque for centrifugal fans and water pumps.
5. **On-Load Transformer Tap Changers:** Thyristorized tap changers switch transformer winding taps under full load without mechanical arcing, contact wear, or line interruption.
6. **Static VAR Compensators (SVC):** Thyristor-Controlled Reactors (TCR) dynamically absorb variable inductive reactive power (Q) to stabilize grid voltages.

Single-Phase Switch-Mode DC-AC Voltage Source Inverters (VSI)

Fundamental Operating Principles and Bridge Topologies

An **Inverter** is a power electronic converter that transforms direct current (DC) electrical energy into alternating current (AC) electrical energy at a desired, precisely regulated voltage magnitude and frequency:

$$\text{DC Input } (V_{dc}) \xrightarrow{\text{Inverter}} \text{AC Output } (v_o(t),\, f_o) \tag{22}$$

In a **Voltage Source Inverter (VSI)**, the DC input source is modeled as an ideal voltage source with negligible internal impedance (stiff DC bus supported by a large electrolytic DC-link capacitor bank). The output voltage waveform is strictly determined by the switching states of the converter power semiconductors, while the output current waveform is governed by the connected load impedance.

Single-Phase Half-Bridge Inverter Configuration

The single-phase half-bridge topology (Figure 3) utilizes two fully controlled semiconductor switches (S_1, S_2) and a three-wire split DC supply with center-tap O (established by two identical series capacitors $C_1 = C_2$, each holding $V_{dc}/2$). Two anti-parallel freewheeling diodes (D_1, D_2) provide a continuous path for inductive load currents.

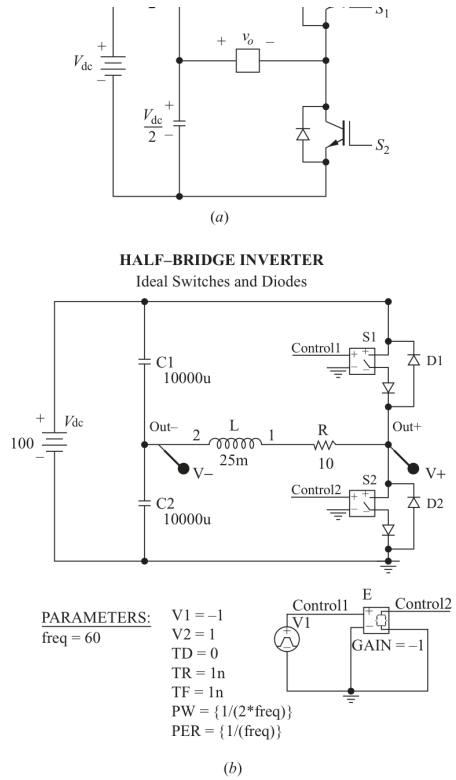


Figure 8-8 (*a*) A half-bridge inverter using IGBTs. The output is $\pm V_{dc}$; (*b*) A PSpice implementation using voltage-controlled switches and diodes.

Figure 3: Single-phase half-bridge voltage source inverter: (a) Power circuit diagram with split DC source $V_{dc}/2$; (b) Square-wave output voltage $v_o(t)$ and inductive load current $i_o(t)$ waveforms showing conduction intervals of switches and freewheeling feedback diodes. (Source: Daniel W. Hart, Fig. 8-1)

- **Interval 1 ($0 \le t < T/2$):** Switch S_1 is turned ON. Load terminal A is connected to the positive DC rail ($+V_{dc}/2$ relative to center tap O):

$$v_o(t) = v_{AO}(t) = +\frac{V_{dc}}{2} \tag{23}$$

- **Interval 2 ($T/2 \leq t < T$):** Switch S_1 is turned OFF, and S_2 is turned ON. Load terminal A is connected to the negative DC rail ($-V_{dc}/2$):

$$v_o(t) = v_{AO}(t) = -\frac{V_{dc}}{2} \tag{24}$$

- **Freewheeling Diode Operation with Inductive Load:** If S_1 turns OFF while positive load current $i_o > 0$ is still flowing, inductor energy forces diode D_2 into conduction, clamping $v_o = -V_{dc}/2$ and feeding reactive energy back into the lower DC capacitor C_2 until i_o decays to zero!

- **RMS Output Voltage:**

$$V_{or} = \sqrt{\frac{1}{T}\left[\int_0^{T/2}\left(+\frac{V_{dc}}{2}\right)^2 dt + \int_{T/2}^{T}\left(-\frac{V_{dc}}{2}\right)^2 dt\right]} = \frac{V_{dc}}{2} \tag{25}$$

- **Major Limitations of the Half-Bridge Inverter:**

  1. Requires a split three-wire DC supply or large center-tapped balancing capacitors.
  2. DC bus voltage utilization is low: the maximum peak-to-peak output voltage is only V_{dc}, yielding a fundamental peak of only $\hat{V}_{o1} = \frac{2}{\pi}V_{dc} \approx 0.637V_{dc}$.
  3. Peak blocking voltage across the OFF-state switch equals the full DC supply $V_{dc} = 2 \times (V_{dc}/2)$, requiring devices rated for twice the output voltage!

Single-Phase Full-Bridge (H-Bridge) Inverter Configuration

The full-bridge inverter (Figure 4) eliminates the center-tap requirement by employing four controlled semiconductor switches (S_1, S_2, S_3, S_4) arranged in two phase legs (A and B) across a single DC source V_{dc}.

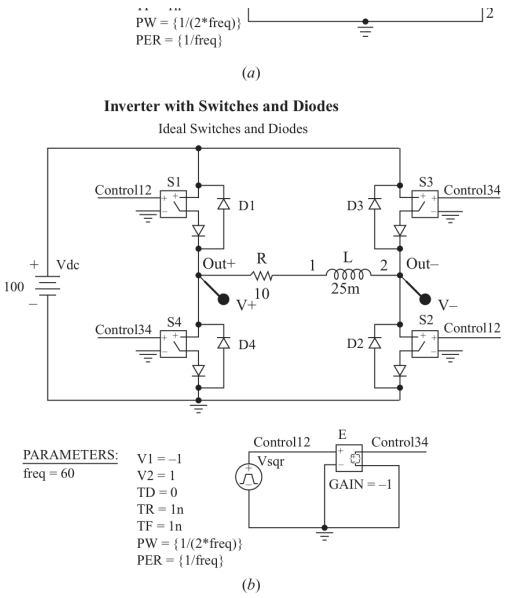


Figure 8-4 (*a*) Square-wave inverter simulation using an ideal source; (*b*) Square-wave inverter using switches and diodes.

Figure 4: Single-phase full-bridge (H-bridge) inverter: (a) Power circuit diagram; (b) Square-wave output voltage $v_o(t) = v_A - v_B$ alternating between $+V_{dc}$ and $-V_{dc}$, and inductive load current $i_o(t)$. (Source: Daniel W. Hart, Fig. 8-4)

- **Diagonal Pair Conduction:**

- When switches (S_1, S_2) are gated ON: $v_{Ao} = +V_{dc}/2$ and $v_{Bo} = -V_{dc}/2$, producing output voltage:

$$v_o(t) = v_{AB}(t) = v_{Ao} - v_{Bo} = +V_{dc} \tag{26}$$

- When switches (S_3, S_4) are gated ON: $v_{Ao} = -V_{dc}/2$ and $v_{Bo} = +V_{dc}/2$, producing output voltage:

$$v_o(t) = v_{AB}(t) = -V_{dc} \tag{27}$$

- **RMS Output Voltage:**

$$V_{or} = \sqrt{\frac{1}{T}\int_0^T (\pm V_{dc})^2 dt} = V_{dc} \tag{28}$$

The full-bridge produces **twice** the output voltage of the half-bridge for the same DC link voltage, yielding four times the output power into a given load resistance!

Fourier Analysis and Harmonic Spectrum of Square-Wave Inverters

The square-wave output voltage $v_o(t)$ of a full-bridge inverter alternating between $+V_{dc}$ and $-V_{dc}$ possesses **odd half-wave symmetry**:

$$v_o(\omega t) = -v_o(\omega t + \pi) \quad \text{and} \quad v_o(-\omega t) = -v_o(\omega t) \tag{29}$$

Consequently, all DC offset and all even harmonics are identically zero ($a_0 = 0$, $a_n = 0$, $b_n = 0$ for even n). The Fourier series contains only odd sine harmonics:

$$b_n = \frac{2}{\pi}\int_0^{\pi} V_{dc}\sin(n\omega t)d(\omega t) = \frac{2V_{dc}}{n\pi}\left[-\cos(n\omega t)\right]_0^{\pi} = \frac{2V_{dc}}{n\pi}\left[1-(-1)^n\right] \tag{30}$$

For odd n ($n = 1, 3, 5, 7, \ldots$), $1-(-1)^n = 2$:

Fourier Series of Single-Phase Full-Bridge Square-Wave Inverter

$$v_o(t) = \sum_{n=1,3,5,\ldots}^{\infty} \frac{4V_{dc}}{n\pi}\sin(n\omega t) = \frac{4V_{dc}}{\pi}\sin(\omega t) + \frac{4V_{dc}}{3\pi}\sin(3\omega t) + \frac{4V_{dc}}{5\pi}\sin(5\omega t) + \ldots \tag{31}$$

Key Harmonic Metrics

- **Peak Amplitude of Fundamental:** $\hat{V}_{o1} = \frac{4V_{dc}}{\pi} \approx 1.273V_{dc}$.
- **RMS Value of Fundamental:** $V_{o1(rms)} = \frac{\hat{V}_{o1}}{\sqrt{2}} = \frac{4V_{dc}}{\sqrt{2}\pi} = \frac{2\sqrt{2}}{\pi}V_{dc} \approx 0.9003V_{dc}$.
- **Harmonic Amplitudes:** The amplitude of the n-th harmonic decreases inversely with harmonic order:

$$\hat{V}_{on} = \frac{\hat{V}_{o1}}{n} \implies \frac{V_{on}}{V_{o1}} = \frac{1}{n} \tag{32}$$

Specifically: 3rd harmonic is 33.3% of fundamental, 5th harmonic is 20%, 7th harmonic is 14.3%.

- **Total Harmonic Distortion (THD) of Voltage:**

$$\text{THD}_v = \frac{\sqrt{V_{or}^2 - V_{o1(rms)}^2}}{V_{o1(rms)}} = \frac{\sqrt{V_{dc}^2 - \left(\frac{2\sqrt{2}}{\pi}V_{dc}\right)^2}}{\frac{2\sqrt{2}}{\pi}V_{dc}} = \sqrt{\frac{\pi^2}{8} - 1} \approx \sqrt{1.2337 - 1} = 0.4834 \implies 48.34\% \tag{33}$$

A total harmonic distortion of 48.34% is severe and unacceptable for sensitive AC loads or utility grid interconnection, motivating the necessity of Pulse Width Modulation (PWM)!

Sinusoidal Pulse Width Modulation (SPWM)

In **Sinusoidal Pulse Width Modulation (SPWM)**, the inverter switches are turned ON and OFF multiple times per half-cycle at a high switching frequency. The widths of the gating pulses are modulated proportionally to the instantaneous amplitude of a sinusoidal reference wave.

Fundamental Definitions and Modulation Parameters

1. **Modulating / Reference Wave (v_{control} or v_{sine}):** A low-frequency sinusoidal voltage whose frequency f_o defines the fundamental output frequency of the inverter, and whose peak amplitude $\hat{V}_{\text{control}}$ dictates the output voltage magnitude.
2. **Carrier Wave (v_{tri}):** A high-frequency symmetrical triangular waveform of frequency f_s (or f_c) and peak amplitude $\hat{V}_{\text{tri}}$.
3. **Amplitude Modulation Index (m_a):**

Amplitude Modulation Index

$$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}} \tag{34}$$

m_a governs the amplitude of the fundamental output voltage component.

4. **Frequency Modulation Ratio (m_f):**

Frequency Modulation Ratio

$$m_f = \frac{f_s}{f_o} = \frac{f_{\text{carrier}}}{f_{\text{reference}}} \tag{35}$$

m_f determines the frequency at which switching harmonics appear in the output spectrum.

PWM with Bipolar Voltage Switching

In bipolar voltage switching, the diagonally opposite switch pairs (S_1, S_2) and (S_3, S_4) in the full bridge are switched simultaneously in complementary fashion:

- When $v_{\text{control}} > v_{\text{tri}}$: Switches S_1 and S_2 are ON $\implies v_o(t) = +V_{dc}$.
- When $v_{\text{control}} < v_{\text{tri}}$: Switches S_3 and S_4 are ON $\implies v_o(t) = -V_{dc}$.

Because the output voltage transitions directly between the two extreme polarities ($+V_{dc}$ and $-V_{dc}$), this scheme is termed **bipolar switching**.

Harmonic Characteristics of Bipolar PWM

- **Fundamental Peak Voltage in Linear Range ($m_a \leq 1.0$):**

$$\hat{V}_{o1} = m_a V_{dc} \implies V_{o1(rms)} = \frac{m_a V_{dc}}{\sqrt{2}} \approx 0.707 m_a V_{dc} \tag{36}$$

The fundamental output voltage is strictly linearly proportional to the modulation index m_a.

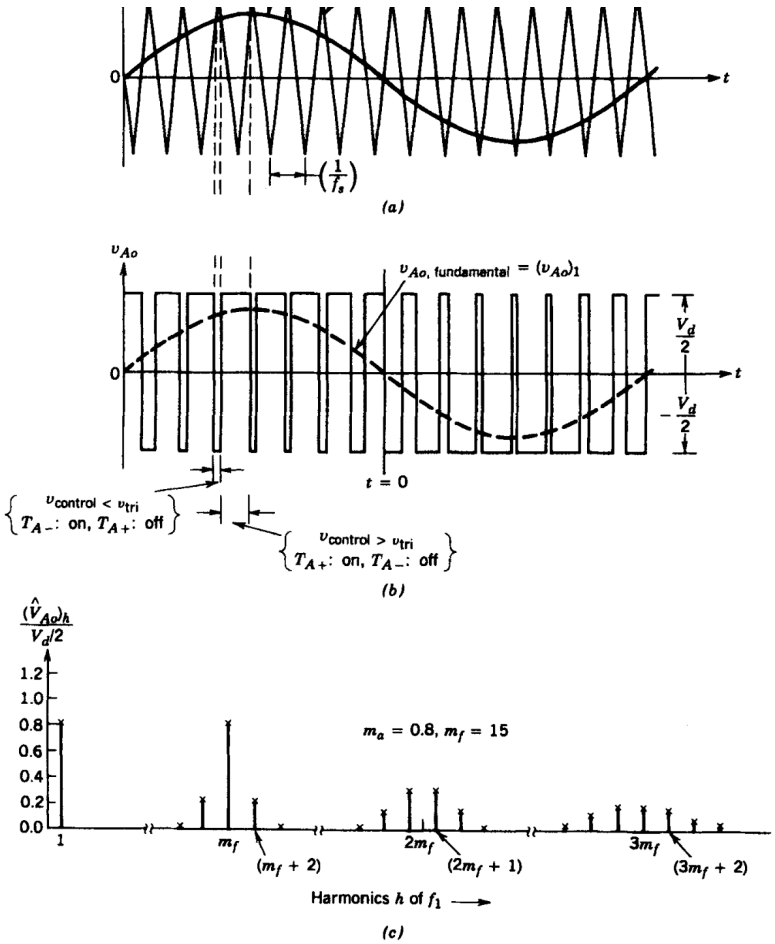


Figure 8-5 Pulse-width modulation.

The frequency modulation ratio m_f is defined as

$$m_f = \frac{f_s}{f_1} \tag{8-2}$$

In the inverter of Fig. 8-4*b*, the switches T_{A+} and T_{A-} are controlled based on the comparison of $v_{control}$ and v_{tri}, and the following output voltage results, independent of the direction of i_o:

$$v_{control} > v_{tri}, \qquad T_{A+} \text{ is on}, \qquad v_{Ao} = \tfrac{1}{2}V_d \tag{8-3}$$

or

Figure 5: Pulse-width modulation with bipolar voltage switching: (a) Comparison of sinusoidal reference v_{control} and triangular carrier v_{tri}; (b) Switched pole voltages v_{Ao} and v_{Bo}; (c) Bipolar output voltage $v_o(t)$ alternating between $+V_{dc}$ and $-V_{dc}$; (d) Harmonic spectrum showing fundamental at f_1 and harmonic sidebands centered around $m_f, 2m_f, 3m_f$. (Source: Ned Mohan, Fig. 8-5)

- **Harmonic Sidebands:** Harmonics appear as sidebands clustered around the switching frequency and its integer multiples:

$$f_h = (j \cdot m_f \pm k)\, f_o \tag{37}$$

  where the harmonic order $h = j \cdot m_f \pm k$.

- **Selection Rules for m_f:**
  - m_f should be chosen as an **odd integer**. This establishes **odd half-wave symmetry** ($v_o(\omega t + \pi) = -v_o(\omega t)$), which completely eliminates all even harmonics from the output spectrum!
  - If m_f is an odd integer, the dominant harmonics occur at m_f and $m_f \pm 2$.
  - Synchronous PWM (m_f = integer) is essential for $m_f < 21$ to prevent sub-harmonic frequencies. For $m_f > 21$, asynchronous PWM can be utilized.

PWM with Unipolar Voltage Switching

In unipolar voltage switching, the two inverter legs are controlled independently by comparing the carrier wave v_{tri} with two reference sine waves that are $180°$ out of phase:

- **Leg A Control:** v_{control} is compared with v_{tri}:
  - If $v_{\text{control}} > v_{\text{tri}}$: Switch S_1 is ON ($v_{Ao} = +V_{dc}/2$).
  - If $v_{\text{control}} < v_{\text{tri}}$: Switch S_4 is ON ($v_{Ao} = -V_{dc}/2$).
- **Leg B Control:** $-v_{\text{control}}$ is compared with v_{tri}:
  - If $-v_{\text{control}} > v_{\text{tri}}$: Switch S_3 is ON ($v_{Bo} = +V_{dc}/2$).
  - If $-v_{\text{control}} < v_{\text{tri}}$: Switch S_2 is ON ($v_{Bo} = -V_{dc}/2$).

The Remarkable Advantages of Unipolar PWM

During each switching event, the output voltage transitions between **three voltage levels**: $+V_{dc}$, 0, and $-V_{dc}$:

- During the positive half-cycle of v_{control}, $v_o(t)$ switches between 0 and $+V_{dc}$.
- During the negative half-cycle of v_{control}, $v_o(t)$ switches between 0 and $-V_{dc}$.

Why Unipolar Switching Outperforms Bipolar Switching

1. **Doubling of Effective Switching Frequency:** The output voltage pulses appear at **twice the carrier frequency** ($2f_s$). The fundamental output voltage equation remains identical:

$$\hat{V}_{o1} = m_a V_{dc} \quad (0 \le m_a \le 1.0) \tag{38}$$

2. **Cancellation of Harmonics at m_f:** Harmonics centered around $m_f, 3m_f, 5m_f, \ldots$ are completely cancelled out! The lowest-order harmonic sidebands appear centered around $2m_f$:

$$h = 2m_f \pm 1, \quad 2m_f \pm 3 \tag{39}$$

3. **Dramatically Smaller Output Filter:** Because the lowest harmonic frequency is doubled ($2f_s$), the required inductance and capacitance of the output low-pass filter are cut in half, slashing converter weight, volume, and cost!
4. **Reduced Current Ripple:** Load current ripple is substantially smaller compared to bipolar PWM for identical device switching frequencies.

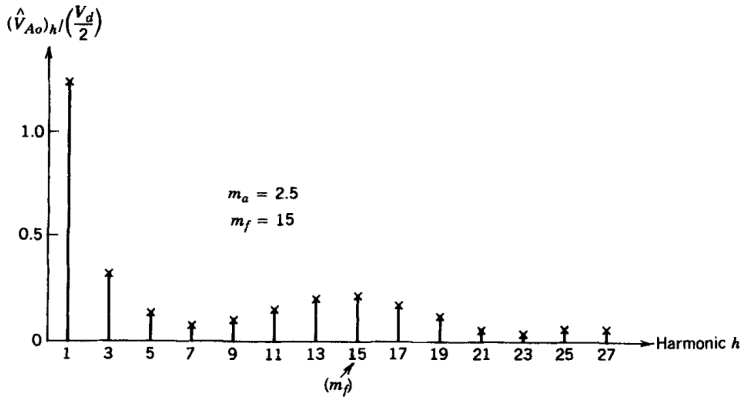


Figure 8-7 Harmonics due to overmodulation; drawn for $m_a = 2.5$ and $m_f = 15$.

overmodulation. More significantly, with overmodulation, the amplitude of the fundamental-frequency component does not vary linearly with the amplitude modulation ratio m_a. Figure 8-8 shows the normalized peak amplitude of the fundamental-frequency component $(\hat{V}_{Ao})_1/\frac{1}{2}V_d$ as a function of the amplitude modulation ratio m_a. Even at reasonably large values of m_f, $(\hat{V}_{Ao})_1/\frac{1}{2}V_d$ depends on m_f in the overmodulation region. This is contrary to the linear range ($m_a \leq 1.0$) where $(\hat{V}_{Ao})_1/\frac{1}{2}V_d$ varies linearly with m_a, almost independent of m_f (provided $m_f > 9$).

With overmodulation regardless of the value of m_f, it is recommended that a synchronous PWM operation be used, thus meeting the requirements indicated previously for a small value of m_f.

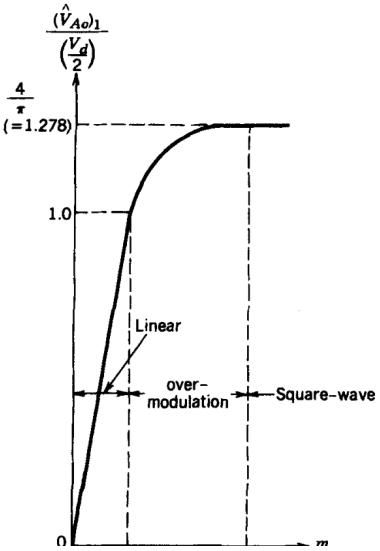


Figure 6: Pulse-width modulation with unipolar voltage switching: (a) Reference waves v_{control} and $-v_{\text{control}}$ compared with carrier v_{tri}; (b) Pole voltages v_{Ao} and v_{Bo}; (c) Unipolar output voltage $v_o(t)$ switching between 0 and $+V_{dc}$ or 0 and $-V_{dc}$; (d) Harmonic spectrum showing the total absence of harmonics around m_f and first harmonic cluster shifted to $2m_f$. (Source: Ned Mohan, Fig. 8-11)

Table 1: Comprehensive Comparison: Bipolar vs. Unipolar PWM Schemes

Feature	Bipolar PWM	Unipolar PWM
Output Voltage Levels	2 levels: $+V_{dc}, -V_{dc}$	3 levels: $+V_{dc}, 0, -V_{dc}$
Leg Control	Legs A and B switched simultaneously	Legs A and B switched independently
Carrier References	Single sine wave v_{control}	Complementary pair: $+v_{\text{control}}, -v_{\text{control}}$
Selection of m_f	m_f must be an odd integer	m_f is typically chosen as an even integer
Effective Output Frequency	f_s (Carrier frequency)	$2f_s$ (Double carrier frequency)
Dominant Harmonics	Clustered around $m_f, 2m_f, 3m_f$	Absent at m_f; first appear around $2m_f$
Output Filter Size	Large	Significantly smaller ($\approx$ 50% reduction)
Switching Stress (dv/dt)	Steps of $2V_{dc}$ across load terminals	Steps of V_{dc} across load terminals
Control Complexity	Minimal	Moderately higher

Operating Modes: Linear Modulation, Overmodulation, and Square-Wave

The relationship between the fundamental output voltage $\hat{V}_{o1}$ and the modulation index m_a spans three distinct operational regimes:

1. **Linear Modulation Region ($0 \leq m_a \leq 1.0$):**
   - The peak of the modulating sine wave is less than or equal to the triangular carrier peak ($\hat{V}_{\text{control}} \leq \hat{V}_{\text{tri}}$).
   - The fundamental peak output voltage varies strictly linearly with m_a:
   $$\hat{V}_{o1} = m_a V_{dc} \tag{40}$$
   - The harmonic frequencies are independent of m_a, and harmonic amplitudes are directly proportional to m_a.
2. **Overmodulation Region ($1.0 < m_a \leq 3.24$):**
   - The reference sine wave amplitude exceeds the carrier peak ($\hat{V}_{\text{control}} > \hat{V}_{\text{tri}}$).
   - For intervals near the sine wave peaks, the reference wave completely exceeds the triangular carrier, resulting in **pulse dropping**—the switch remains continuously ON for multiple carrier cycles without pulsing.
   - The relationship between $\hat{V}_{o1}$ and m_a becomes **non-linear** and begins to saturate.
   - Low-order harmonics (3rd, 5th, 7th) re-appear directly in the output voltage spectrum, increasing harmonic distortion. However, overmodulation is widely utilized in industrial AC motor drives to extract maximum fundamental voltage during high-speed operation!
3. **Square-Wave Operation ($m_a > 3.24$):**
   - As m_a becomes large ($m_a > 3.24$), the PWM pulses merge completely into a single square wave per half-cycle.
   - The inverter operates in pure square-wave mode, and the fundamental peak saturates at its absolute theoretical maximum:
   $$\hat{V}_{o1,\text{max}} = \frac{4V_{dc}}{\pi} \approx 1.273 V_{dc} \tag{41}$$
   - In this regime, the output voltage is completely independent of m_a and can only be regulated by varying the input DC link voltage V_{dc}.

The Critical Need for Blanking Time (Dead-Time)

In practical bridge inverters, power semiconductor switches (IGBTs, MOSFETs) do not turn OFF instantaneously due to stored minority carrier recombination and gate charge discharging times ($t_{off} > t_{on}$).

8-4-1 PWM IN THREE-PHASE VOLTAGE SOURCE INVERTERS

Similar to the single-phase inverters, the objective in pulse-width-modulated three-phase inverters is to shape and control the three-phase output voltages in magnitude and frequency with an essentially constant input voltage V_d. To obtain balanced three-phase output voltages in a three-phase PWM inverter, the same triangular voltage waveform is compared with three sinusoidal control voltages that are 120° out of phase, as shown in Fig. 8-22*a* (which is drawn for $m_f = 15$).

It should also be noted from Fig. 8-22*b* that an identical amount of average dc component is present in the output voltages v_{AN} and v_{BN}, which are measured with respect to the negative dc bus. These dc components are canceled out in the line-to-line voltages, for example in v_{AB} shown in Fig. 8-22*b*. This is similar to what happens in a single-phase full-bridge inverter utilizing a PWM switching.

In the three-phase inverters, only the harmonics in the line-to-line voltages are of concern. The harmonics in the output of any one of the legs, for example v_{AN} in Fig. 8-22*b*, are identical to the harmonics in v_{Ao} in Fig. 8-5, where only the odd harmonics exist as sidebands, centered around m_f and its multiples, provided m_f is odd. Only considering the harmonic at m_f (the same applies to its odd multiples), the phase difference between the m_f harmonic in v_{AN} and v_{BN} is (120 m_f)°. This phase difference will be equivalent to zero (a multiple of 360°) if m_f is odd and a multiple of 3. As a consequence, the harmonic at m_f is suppressed in the line-to-line voltage v_{AB}. The same argument applies in the suppression of harmonics at the odd multiples of m_f if m_f is chosen to be an odd multiple of 3 (where the reason for choosing m_f to be an odd multiple of 3 is to keep m_f odd and, hence, eliminate even harmonics). Thus, some of the dominant harmonics in the one-leg inverter can be eliminated from the line-to-line voltage of a three-phase inverter.

PWM considerations are summarized as follows:

1. *For low values of* m_f, to eliminate the even harmonics, a synchronized PWM should be used and m_f should be an odd integer. Moreover, m_f should be a multiple of 3 to cancel out the most dominant harmonics in the line-to-line voltage.
2. *For large values of* m_f, the comments in Section 8-2-1-2 for a single-phase PWM

Figure 7: Effect of blanking time (dead time) in a bridge inverter: (a) Inverter leg showing upper and lower switches and freewheeling diodes; (b) Gate signals with blanking time t_Δ; (c) Resulting pole voltage waveforms showing voltage reduction for positive current and voltage increase for negative current. (Source: Ned Mohan, Fig. 8-31)

The Shoot-Through Catastrophe

If the turn-ON gate command is applied to the upper switch (S_1) at the exact instant the turn-OFF command is sent to the lower switch (S_4) in the same leg, both switches will be simultaneously ON for the turn-off duration t_{off}. This creates a zero-impedance **shoot-through dead short circuit directly across the DC bus**:

$$I_{\text{fault}} = \frac{V_{dc}}{R_{\text{parasitic}}} \rightarrow \text{Thousands of Amperes} \implies \text{Explosive device destruction!} \tag{42}$$

Implementation of Blanking Time (t_d or t_Δ)

To prevent shoot-through, a deliberate hardware/software delay called **blanking time** (or **dead-time**, t_d) is inserted between the turn-OFF command of one switch and the turn-ON command of the complementary switch in the same leg:

$$t_d > t_{off,\text{max}} + t_{\text{driver delay mismatch}} \tag{43}$$

Typical values: $1.0 - 3.0\,\mu$s for IGBTs; $100 - 500$ ns for silicon MOSFETs; < 50 ns for GaN HEMTs.

Adverse Effects and Voltage Penalty Caused by Blanking Time

During the blanking interval t_d, both controlled switches in the leg are commanded OFF. The actual pole voltage is determined entirely by the **direction of the load current** flowing through the freewheeling diodes:

- **When $i_o > 0$:** The current must flow through lower diode D_4. Even though S_1 was commanded ON, the pole voltage is clamped to 0 V instead of $+V_{dc}$. The pulse width is reduced by t_d, causing a loss of output voltage!
- **When $i_o < 0$:** The current flows through upper diode D_1, clamping the pole voltage to $+V_{dc}$ instead of 0 V. The pulse width is increased by t_d.
- **Fundamental Voltage Drop:** Averaging this error over one fundamental cycle results in an average fundamental voltage reduction of:

$$\Delta V_o = \frac{4t_d}{T_s} V_{dc} = 4 f_s t_d V_{dc} \tag{44}$$

- **Adverse Consequences:**
  1. Causes a noticeable reduction in fundamental output voltage, especially prominent at low modulation indices.
  2. Generates severe **low-order harmonics** (5th, 7th, 11th) that are in-phase with current zero-crossings, producing torque pulsations in AC motors.
  3. Introduces waveform distortion and zero-crossing deadbands in precision motor drives.

Three-Phase Pulse Width Modulated Voltage Source Inverters

Three-Phase Bridge Inverter Topology and Conduction Mechanics

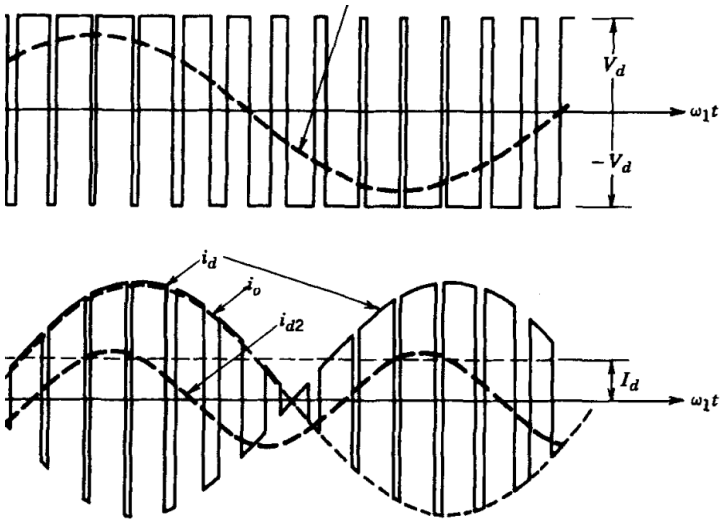
The three-phase voltage source inverter (Figure 8) is the universal industry-standard power conversion topology for variable-frequency AC motor drives (VFDs), grid-tied solar photovoltaic central inverters, wind turbine grid converters, and electric vehicle traction drives (50 kW $- 300$ kW).

The topology comprises six controlled switches (S_1 through S_6) arranged in three inverter legs (A, B, C) across a common DC link voltage V_{dc}, with six anti-parallel freewheeling diodes (D_1 through D_6).

Switch Numbering and Firing Sequence

The switches are traditionally numbered according to their natural firing sequence in a balanced three-phase cycle:

- Upper switches: S_1 (Phase A), S_3 (Phase B), S_5 (Phase C).
- Lower switches: S_4 (Phase A), S_6 (Phase B), S_2 (Phase C).
- Firing sequence: $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5 \rightarrow S_6$, spaced by intervals of 60° ($\pi/3$).



3-14 The dc-side current in a single-phase inverter with PWM voltage switching.

ractical systems, the previous assumption of a constant dc voltage as the input to

Figure 8: Three-phase switch-mode voltage source inverter power circuit: Three phase legs (A, B, C) feeding a three-phase star (Y) or delta (Δ) connected AC load from a stiff DC link capacitor bank. (Source: Ned Mohan, Fig. 8-21)

Sinusoidal PWM in Three-Phase VSIs

In three-phase SPWM, three balanced sinusoidal reference voltages displaced by 120° are compared against a common high-frequency triangular carrier wave v_{tri}:

$$\begin{aligned} v_{\text{control},A} &= \hat{V}_{\text{control}} \sin(\omega t) \\ v_{\text{control},B} &= \hat{V}_{\text{control}} \sin(\omega t - 120^\circ) \\ v_{\text{control},C} &= \hat{V}_{\text{control}} \sin(\omega t - 240^\circ) = \hat{V}_{\text{control}} \sin(\omega t + 120^\circ) \end{aligned} \tag{45}$$

Pole Voltages and Line-to-Line Voltages

Each leg operates identically to a single-phase half-bridge pole relative to the fictitious DC-bus midpoint o:

$$v_{Ao}(t) = \begin{cases} +V_{dc}/2, & \text{if } v_{\text{control},A} > v_{\text{tri}} \\ -V_{dc}/2, & \text{if } v_{\text{control},A} < v_{\text{tri}} \end{cases} \tag{46}$$

The line-to-line output voltages are obtained by taking the difference between individual leg pole voltages:

$$\begin{aligned} v_{AB}(t) &= v_{Ao}(t) - v_{Bo}(t) \\ v_{BC}(t) &= v_{Bo}(t) - v_{Co}(t) \\ v_{CA}(t) &= v_{Co}(t) - v_{Ao}(t) \end{aligned} \tag{47}$$

Cancellation of Triplen Harmonics in Three-Phase PWM

A fundamental property of three-phase balanced systems is that **all triplen harmonics ($n = 3, 9, 15, 21, \ldots$) are identical in magnitude and in-phase across all three phase legs**:

$$v_{Ao,3}(t) = v_{Bo,3}(t) = v_{Co,3}(t) \tag{48}$$

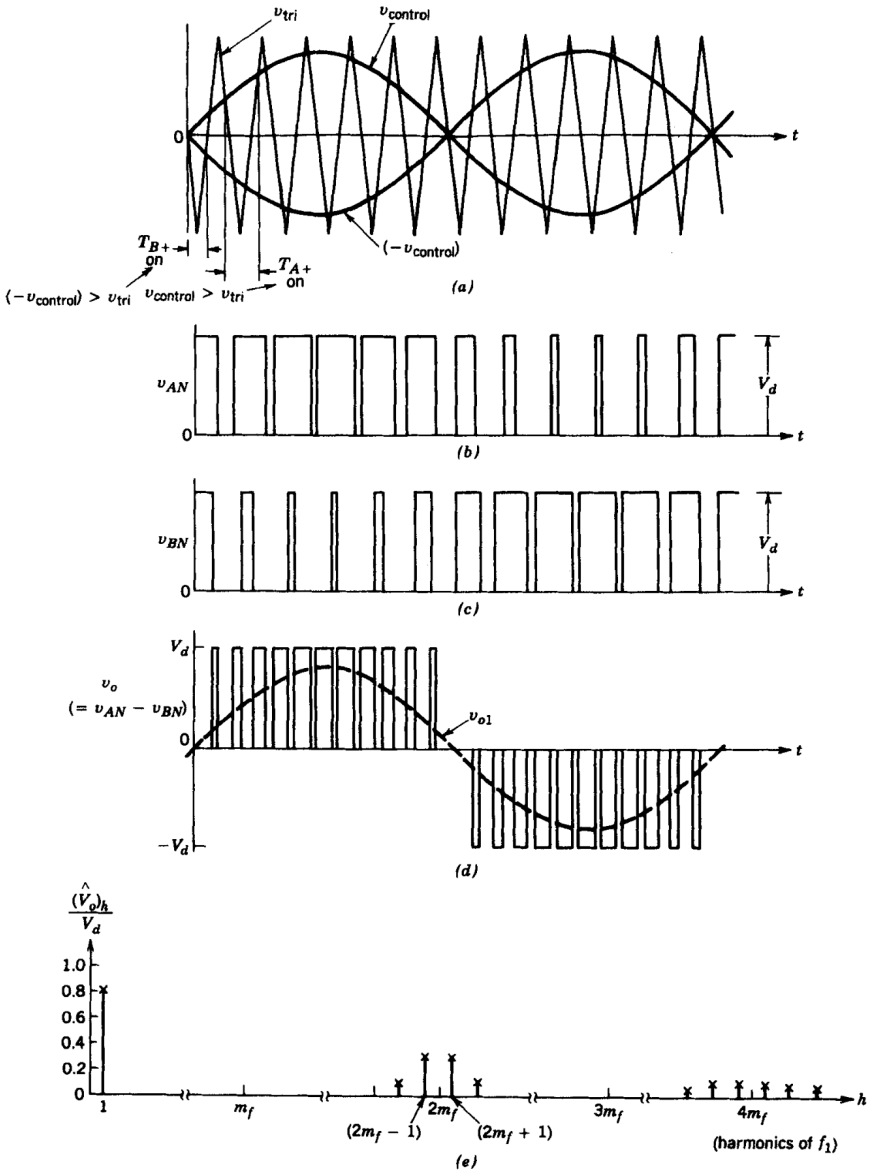


Figure 8-15 PWM with unipolar voltage switching (single phase).

Figure 9: Three-phase PWM waveforms and harmonic spectrum: (a) Modulation references and carrier; (b) Pole voltages v_{Ao} and v_{Bo}; (c) Line-to-line output voltage $v_{AB}(t) = v_{Ao} - v_{Bo}$; (d) Harmonic spectrum of line-to-line voltage showing total cancellation of all triplen harmonics ($3rd, 9th, 15th, \ldots$). (Source: Ned Mohan, Fig. 8-22)

Taking the line-to-line difference completely cancels all triplen harmonics:

$$v_{AB,3}(t) = v_{Ao,3}(t) - v_{Bo,3}(t) \equiv 0! \tag{49}$$

Fundamental Voltage Relationships in 3-Phase SPWM

In the linear modulation range ($0 \leq m_a \leq 1.0$):

1. **Peak Fundamental Pole-to-Midpoint Voltage:**

$$\hat{V}_{Ao1} = m_a \frac{V_{dc}}{2} \tag{50}$$

2. **Peak Fundamental Line-to-Line Voltage:**

$$\hat{V}_{LL1} = \sqrt{3}\hat{V}_{Ao1} = \sqrt{3} m_a \frac{V_{dc}}{2} = \frac{\sqrt{3}}{2} m_a V_{dc} \approx 0.866 m_a V_{dc} \tag{51}$$

3. **RMS Fundamental Line-to-Line Voltage:**

$$V_{LL1(rms)} = \frac{\hat{V}_{LL1}}{\sqrt{2}} = \frac{\sqrt{3}}{2\sqrt{2}} m_a V_{dc} = \frac{\sqrt{6}}{4} m_a V_{dc} \approx 0.6124 m_a V_{dc} \tag{52}$$

4. **Maximum Linear Line-to-Line RMS Voltage ($m_a = 1.0$):**

$$V_{LL1(rms),\text{max}} = 0.6124 V_{dc} \tag{53}$$

Three-Phase Frequency Modulation Ratio (m_f) Selection Criteria

To ensure optimum harmonic elimination in three-phase inverters, m_f must adhere to two strict design criteria:

1. m_f must be an **odd integer**: Ensures half-wave symmetry, completely eliminating all even harmonics.
2. m_f must be an **odd multiple of 3** ($m_f = 9, 15, 21, 27, 33, \ldots$):
   - Forces the dominant carrier-frequency harmonic cluster to appear at m_f and its multiples.
   - Because m_f is a multiple of 3, this dominant harmonic cluster becomes a **triplen harmonic**, which is **completely cancelled out from the line-to-line voltage**!
   - The lowest harmonic sidebands remaining in the line-to-line output voltage are pushed up to $(m_f \pm 2)$ and $(m_f \pm 4)$.

Overmodulation and Square-Wave Operation in Three-Phase Inverters

Overmodulation Region ($1.0 < m_a \leq 3.24$)

When m_a exceeds 1.0, the peak reference wave exceeds the carrier amplitude. The fundamental line-to-line output voltage increases beyond $0.6124 V_{dc}$, but non-linear saturation occurs and low-order non-triplen harmonics (5th, 7th, 11th, 13th) appear in the output.

Square-Wave Operation ($m_a > 3.24$ / Six-Step Inverter)

In the extreme limit, the inverter operates as a classical **six-step (square-wave) inverter**, where each switch conducts for $180°$ per cycle.

- The line-to-line voltage is a six-step waveform alternating between $+V_{dc}, 0$, and $-V_{dc}$.
- The Fourier series of the line-to-line voltage is:

$$v_{AB}(t) = \sum_{n=1,5,7,11,13,\ldots}^{\infty} \frac{4V_{dc}}{n\pi} \cos\left(\frac{n\pi}{6}\right) \sin(n\omega t) \tag{54}$$

  Notice that for all triplen harmonics ($n = 3, 9, 15, \ldots$), $\cos(n\pi/6) = 0$, confirming complete triplen elimination!

- **Peak Fundamental Line-to-Line Voltage:**

$$\hat{V}_{LL1,\text{square}} = \frac{4V_{dc}}{\pi} \cos\left(\frac{\pi}{6}\right) = \frac{4V_{dc}}{\pi}\frac{\sqrt{3}}{2} = \frac{2\sqrt{3}}{\pi}V_{dc} \approx 1.1026V_{dc} \tag{55}$$

- **RMS Fundamental Line-to-Line Voltage in Square-Wave Mode:**

$$V_{LL1(rms),\text{square}} = \frac{\hat{V}_{LL1,\text{square}}}{\sqrt{2}} = \frac{\sqrt{6}}{\pi}V_{dc} \approx 0.7797V_{dc} \tag{56}$$

- **Total RMS Value of Line-to-Line Voltage:**

$$V_{LL(rms)} = \sqrt{\frac{1}{\pi}\int_0^{2\pi/3} V_{dc}^2 d(\omega t)} = V_{dc}\sqrt{\frac{2}{3}} \approx 0.8165V_{dc} \tag{57}$$

Third Harmonic Injection PWM (THIPWM)

In standard sinusoidal PWM, the maximum linear fundamental line-to-line voltage is limited to $\hat{V}_{LL1} = \frac{\sqrt{3}}{2}V_{dc} \approx 0.866V_{dc}$ because the peak of each phase reference sine wave cannot exceed the DC rails ($\pm V_{dc}/2$) without causing overmodulation distortion.

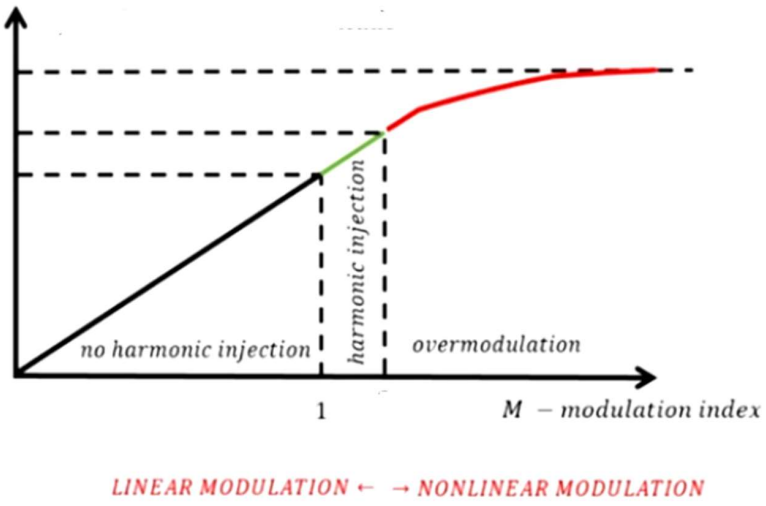


Figure 10: Third Harmonic Injection PWM (THIPWM): Addition of a 3rd harmonic zero-sequence component ($\frac{1}{6}\hat{V}_m \sin 3\omega t$) flattens the reference waveform peak, preventing carrier clipping while boosting DC bus utilization by 15.5%. (Source: CET Lecture Slides)

Mathematical Formulation of THIPWM

By adding a triplen zero-sequence third-harmonic component to the fundamental sinusoidal reference:

$$v_{\text{control},A}(t) = \hat{V}_1 \left[\sin(\omega t) + \frac{1}{6} \sin(3\omega t) \right] \tag{58}$$

The optimal amplitude of the injected third harmonic is exactly $\frac{1}{6} \approx 16.7\%$ of the fundamental amplitude.

Why Third Harmonic Injection Works

Physics of Third Harmonic Injection

1. **Peak Flattening:** At $\omega t = \pi/2$ (90°), where $\sin(\omega t) = 1$, the third harmonic evaluates to $\sin(3 \cdot \pi/2) = \sin(270°) = -1$. The reference voltage becomes:

$$v_{\text{control}}(\pi/2) = \hat{V}_1 \left[1 - \frac{1}{6} \right] = \frac{5}{6} \hat{V}_1 \tag{59}$$

   The peak of the reference waveform is **flattened by a factor of** $\frac{\sqrt{3}}{2} \approx 0.866$!

2. **15.5% Voltage Boost in Linear Range:** Because the peak is flattened, the fundamental reference amplitude $\hat{V}_1$ can be boosted by $\frac{2}{\sqrt{3}} \approx 1.1547$ (+15.47%) before the waveform touches the carrier peak!

$$\hat{V}_{LL1,\text{max (THIPWM)}} = 1.1547 \times 0.866 V_{dc} = 1.000 V_{dc} \tag{60}$$

$$V_{LL1(rms),\text{max (THIPWM)}} = \frac{V_{dc}}{\sqrt{2}} \approx 0.707 V_{dc} \tag{61}$$

   This represents an immense **15.5% increase in output voltage** compared to standard SPWM ($0.707 V_{dc}$ vs $0.612 V_{dc}$) without leaving the linear modulation region!

3. **Zero Distortion Across Load:** Because the three phases are identical and balanced, the injected third harmonic components across phase legs A, B, and C are in-phase and cancel completely in the line-to-line voltage:

$$v_{AB}(t) = v_{Ao}(t) - v_{Bo}(t) = \text{Pure fundamental without any 3rd harmonic!} \tag{62}$$

4. **Equivalence to Space Vector PWM (SVPWM):** Third harmonic injection PWM produces line-to-line voltage waveforms and DC bus utilization virtually identical to Space Vector Modulation (SVM), but with the computational simplicity of carrier-based analog/digital comparators!

Current Source Inverters (CSI) and Detailed Comparison with VSI

Operating Principle of Single-Phase Current Source Inverters (CSI)

In contrast to a Voltage Source Inverter (which is fed from a stiff, low-impedance DC voltage source), a **Current Source Inverter (CSI)** is fed from an adjustable direct current source possessing **high internal impedance**. In practice, this is realized by placing a large smoothing choke (L_{dc}) in series with a controlled DC power supply or phase-controlled thyristor rectifier.

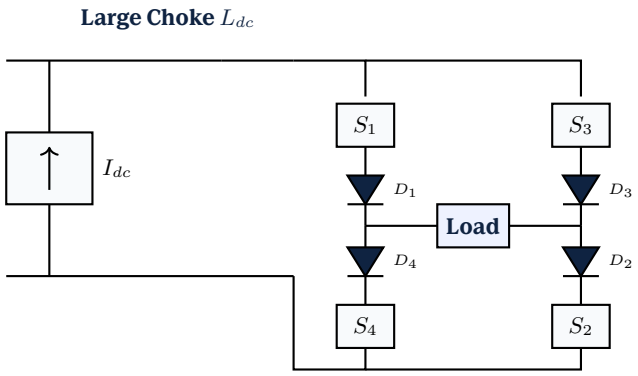


Figure 11: Single-phase IGBT-based Current Source Inverter (CSI) with series blocking diodes and massive DC smoothing inductor L_{dc}.

Circuit Characteristics of the CSI

1. **Constant Input Current (I_{dc}):** Due to the large inductance L_{dc}, the DC link current cannot change instantaneously:
$$i_{dc}(t) \approx I_{dc} = \text{constant (ripple-free)} \tag{63}$$

2. **Square-Wave Output Current Waveform:** Gating switch pair (S_1, S_2) directs $+I_{dc}$ through the load, while gating (S_3, S_4) directs $-I_{dc}$ through the load. The output current $i_o(t)$ is therefore a **clean square wave alternating between $+I_{dc}$ and $-I_{dc}$.**

3. **Need for Series Blocking Diodes ($D_1 - D_4$):**
   - Standard power MOSFETs and IGBTs have anti-parallel body diodes or low reverse-blocking voltage ratings (< 20 V).
   - In a CSI, the load voltage is alternating AC, meaning the switches must withstand **large reverse voltages**.
   - Fast-recovery diodes are placed strictly in **series** with each controlled switch to provide the necessary reverse voltage blocking capability!

4. **Inherent Short-Circuit Immunity:** In a VSI, accidental simultaneous conduction of upper and lower switches causes an explosive shoot-through short circuit. In sharp contrast, in a CSI, **simultaneous conduction of upper and lower switches merely circulates the constant DC current I_{dc} without any current spike!**

5. **Open-Circuit Danger:** In a CSI, an open circuit in the load or DC path is catastrophic ($L\frac{di}{dt} \rightarrow \infty$, generating thousands of volts of inductive overvoltage arc breakdown). An overlap period is mandatory during commutation!

Comprehensive Technical Comparison: VSI vs. CSI

The comparison between Voltage Source Inverters and Current Source Inverters is a quintessential KTU university examination question (often awarded 6 to 9 marks). Table 2 provides an exhaustive, multi-dimensional comparison.

Utility Interface, Harmonics, IEEE 519 Standard, and Active PFC

Table 2: Definitive Technical Comparison: Voltage Source Inverter (VSI) vs. Current Source Inverter (CSI)

Feature / Parameter	Voltage Source Inverter (VSI)	Current Source Inverter (CSI)
DC Source Characteristic	Stiff DC voltage source with very small internal impedance	Stiff DC current source with very high internal impedance
Energy Storage Element	Large DC-link capacitor bank (C_{dc}) across input terminals	Large DC-link smoothing inductor / choke (L_{dc}) in series
Output Voltage Waveform	Controlled square wave or PWM pulsed waveform (independent of load)	Determined entirely by load impedance and current
Output Current Waveform	Determined entirely by the connected load impedance	Pure square wave or PWM current waveform (independent of load)
Freewheeling / Blocking Diodes	Requires freewheeling diodes in **anti-parallel** with switches	Requires blocking diodes in **series** with switches
Regeneration Capability	Requires a four-quadrant dual converter or active front-end rectifier	Natural and simple: just reverse the DC voltage polarity while I_{dc} remains unidirectional
Short-Circuit Behavior	Highly vulnerable: leg shoot-through creates catastrophic short circuit	Inherent short-circuit ride-through; fault current limited by I_{dc}
Open-Circuit Behavior	Safe: open circuit causes no damage	Extremely hazardous: open-circuit produces destructive inductive $L\frac{di}{dt}$ voltage spikes
Switching Overlap Requirements	Strict **blanking time (dead-time)** mandatory to avoid shoot-through	Strict **overlap time** mandatory to avoid open-circuiting the DC choke
Switch Ratings	High current rating, unidirectional voltage blocking	High reverse voltage blocking capability required
Dynamic Response	Extremely fast (voltage switched instantaneously)	Slower due to large magnetic time constant of L_{dc}
Size, Weight & Cost	Compact and lightweight (uses electrolytic/film capacitors)	Bulky, heavy, and expensive due to massive iron-core DC reactor
Primary Industrial Use	General-purpose industrial VFDs, solar inverters, EV traction, UPS	High-power multi-megawatt synchronous motor drives, induction heating

Degradation of the Electric Utility Interface

Conventional line-frequency diode rectifiers and phase-controlled thyristor converters feeding DC filter capacitors draw **highly non-sinusoidal, pulsed input currents** from the electric utility AC distribution grid.

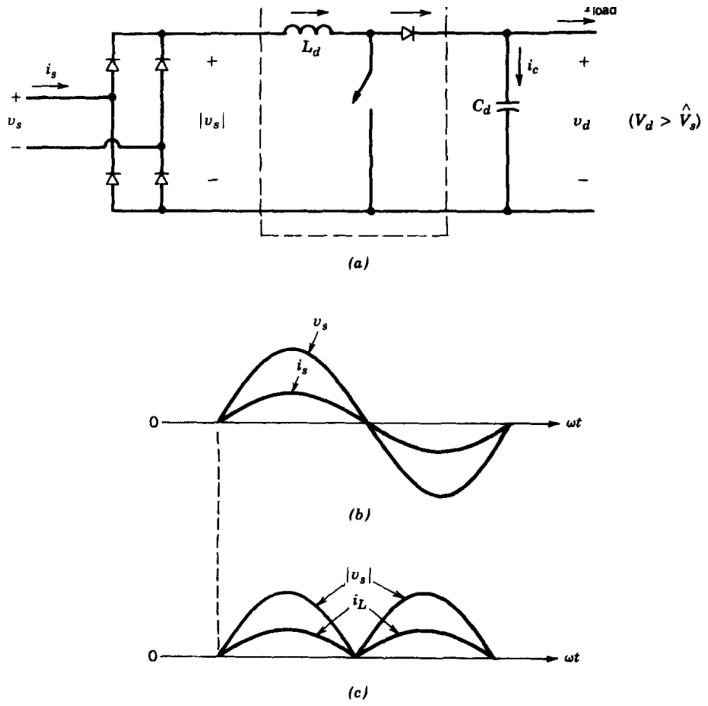


Figure 18-4 Active harmonic filtering: (*a*) step-up converter for current shaping; (*b*) line waveforms; (*c*) v_s and i_L.

Figure 12: Optimizing the utility interface: (a) Active power factor correction (PFC) step-up boost converter shaping the input current into a pure sinusoid in-phase with line voltage; (b) Input line current i_s, switch duty cycle, and high-frequency inductor current i_L. (Source: Ned Mohan, Fig. 18-4)

Harmonic Generation and Detrimental Impacts on Power Systems

1. **Excessive Neutral Current in Three-Phase Four-Wire Systems:** Triplen current harmonics ($3rd, 9th, 15th, \ldots$ generated by single-phase computer SMPS and electronic ballasts do not cancel in the neutral conductor; they add arithmetically in-phase:

$$I_N = \sqrt{I_{N,3}^2 + I_{N,9}^2 + \ldots} \approx 3 \cdot I_{\text{phase},3} \tag{64}$$

   This causes neutral conductors to carry up to $1.73 - 2.0\times$ the phase current, leading to transformer overheating and neutral cable fires!

2. **Transformer and Generator De-rating:** High-frequency harmonic currents induce massive eddy current ($P_{\text{eddy}} \propto f^2$) and stray load losses in magnetic cores, requiring costly K-factor rated transformers.

3. **Voltage Distortion Across the Power Grid:** The harmonic current I_h drawn through the finite utility grid source impedance ($Z_s = R_s + j\omega L_s$) causes voltage drops that distort the supply voltage at the Point of Common Coupling (PCC):

$$V_{h,\text{PCC}} = I_h \cdot Z_{s,h} \tag{65}$$

   This degrades the voltage waveform delivered to neighboring consumers.

4. **Harmonic Resonance with Power Factor Correction Capacitors:** Grid inductances and utility shunt capacitor banks form parallel resonant circuits ($f_{\text{res}} = \frac{1}{2\pi\sqrt{L_{\text{grid}}C_{\text{pfc}}}}$). Harmonic currents near f_{res} excite catastrophic overvoltage resonances, exploding capacitor banks!

Power Factor in Non-Sinusoidal Circuits

When the supply voltage is purely sinusoidal $v_s(t) = \sqrt{2}V_s \sin(\omega t)$ but the input current is distorted by non-linear electronics:

$$i_s(t) = \sqrt{2}I_{s1}\sin(\omega t - \phi_1) + \sum_{h=2}^{\infty}\sqrt{2}I_{sh}\sin(h\omega t - \phi_h) \tag{66}$$

The total RMS current is given by:

$$I_s = \sqrt{I_{s1}^2 + \sum_{h=2}^{\infty} I_{sh}^2} = \sqrt{I_{s1}^2 + I_{\text{dist}}^2} \tag{67}$$

Active Real Power (P) and Apparent Power (S)

Because the cross-product of orthogonal sinusoids of different frequencies integrates to zero over a full cycle, **only the fundamental current component I_{s1} produces average real power**:

$$P = \frac{1}{T}\int_0^T v_s(t)i_s(t)dt = V_s I_{s1}\cos\phi_1 \tag{68}$$

The total apparent power is:

$$S = V_s I_s = V_s\sqrt{I_{s1}^2 + I_{\text{dist}}^2} \tag{69}$$

Displacement Power Factor (DPF) vs. Distortion Power Factor

The true Power Factor (PF) is defined as the ratio of real active power to total apparent power:

$$\text{PF} = \frac{P}{S} = \frac{V_s I_{s1}\cos\phi_1}{V_s I_s} = \left(\frac{I_{s1}}{I_s}\right)\cdot\cos\phi_1 \tag{70}$$

True Power Factor Decomposition

$$\text{PF} = K_{\text{dist}}\cdot\text{DPF} = \left(\frac{I_{s1}}{I_s}\right)\cos\phi_1 \tag{71}$$

where:

- $\text{DPF} = \cos\phi_1$ is the **Displacement Power Factor** (phase displacement of fundamental current relative to voltage).
- $K_{\text{dist}} = \frac{I_{s1}}{I_s}$ is the **Distortion Factor** (purity of the current waveform).

Relationship Between Distortion Factor and Total Harmonic Distortion (THD)

The Total Harmonic Distortion of the current is defined as:

$$\text{THD}_i = \frac{\sqrt{\sum_{h=2}^{\infty} I_{sh}^2}}{I_{s1}} = \frac{\sqrt{I_s^2 - I_{s1}^2}}{I_{s1}} = \sqrt{\left(\frac{I_s}{I_{s1}}\right)^2 - 1} \tag{72}$$

Rearranging gives the direct link between Distortion Factor and THD_i:

$$\frac{I_{s1}}{I_s} = \frac{1}{\sqrt{1 + \text{THD}_i^2}} \tag{73}$$

Substituting into equation (71) yields the foundational formula:

Master Formula: True Power Factor from THD

$$\text{PF} = \frac{\cos\phi_1}{\sqrt{1 + \text{THD}_i^2}} \tag{74}$$

Crucial Exam Insight

Even if the displacement power factor is unity ($\cos\phi_1 = 1.0$), a conventional bridge rectifier with a typical $\text{THD}_i = 80\%$ has a true power factor of only:

$$\text{PF} = \frac{1.0}{\sqrt{1 + (0.80)^2}} = \frac{1.0}{\sqrt{1 + 0.64}} = \frac{1.0}{1.2806} = 0.7808 \implies 78.1\%! \tag{75}$$

This proves that **harmonic distortion alone degrades power factor**, regardless of capacitive/inductive phase shifts!

IEEE 519 Standard for Harmonic Control in Electric Power Systems

The **IEEE 519 Standard** (current revision IEEE 519-2022, "Standard for Harmonic Control in Electric Power Systems") establishes mandatory limits on harmonic distortion at the **Point of Common Coupling (PCC)** between an industrial or commercial customer and the utility transmission/distribution grid.

PCC and Total Demand Distortion (TDD)

IEEE 519 evaluates current distortion using **Total Demand Distortion (TDD)** rather than conventional THD:

$$\text{TDD} = \frac{\sqrt{\sum_{h=2}^{\infty} I_h^2}}{I_L} \tag{76}$$

where I_L is the **maximum demand load current** (fundamental frequency component of the maximum demand occurring over a 15-minute or 30-minute demand interval), while THD is calculated relative to instantaneous fundamental current I_1. Under light load, THD can be artificially astronomical ($> 100\%$) while the actual harmonic Amperes injected into the grid are tiny; TDD prevents this distortion artifact!

IEEE 519 Current Distortion Limits (for Systems Rated 120 V to 69 kV)

The permissible harmonic current limits depend directly on the **Short-Circuit Ratio (SCR)** at the PCC:

$$\text{SCR} = \frac{I_{sc}}{I_L} \tag{77}$$

where I_{sc} is the maximum short-circuit current available at the PCC, and I_L is the maximum fundamental load demand current. A stiffer grid (higher I_{sc}) can absorb more harmonic current without suffering severe voltage distortion.

Table 3: IEEE 519 Current Distortion Limits at the PCC (120 V − 69 kV)

I_{sc}/I_L **Ratio**	$h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	**Max TDD (%)**
$< 20^*$	4.0%	2.0%	1.5%	0.6%	0.3%	**5.0**%
$20 < 50$	7.0%	3.5%	2.5%	1.0%	0.5%	**8.0**%
$50 < 100$	10.0%	4.5%	4.0%	1.5%	0.7%	**12.0**%
$100 < 1000$	12.0%	5.5%	5.0%	2.0%	1.0%	**15.0**%
> 1000	15.0%	7.0%	6.0%	2.5%	1.4%	**20.0**%

*All power generation equipment is restricted to these lowest values regardless of actual I_{sc}/I_L.

Table 4: IEEE 519 Voltage Distortion Limits at the PCC

Bus Voltage at PCC (V)	**Individual Harmonic (%)**	**Total Voltage Distortion THD$_v$ (%)**
$V \leq 1.0$ kV	5.0%	**8.0**%
1.0 kV $< V \leq$ 69 kV	3.0%	**5.0**%
69 kV $< V \leq$ 161 kV	1.5%	**2.5**%
$V >$ 161 kV	1.0%	**1.5**%

Active Shaping of the Input Line Current (Power Factor Correction)

To meet the rigorous harmonic constraints of IEEE 519 and IEC 61000-3-2, modern power supplies employ an **Active Power Factor Correction (PFC) Boost Converter** operating at high switching frequency ($50 - 200$ kHz) directly downstream of the uncontrolled diode bridge.

Circuit Configuration and Dual-Loop Control Mechanics

As depicted in Figures 12 and 13, the active PFC step-up converter consists of a full-wave diode bridge, boost inductor L, power MOSFET/IGBT switch S, fast diode D, and DC bulk capacitor C_d.

The control system employs two cascaded feedback control loops:

1. **Slow Outer Voltage Loop (DC Bus Regulation):**
   - Senses the DC link voltage V_d and compares it to a precise reference V_d^* (typically 390 V − 400 V DC for universal AC mains of 85 V − 265 V).
   - A proportional-integral (PI) voltage controller generates an error voltage signal v_e.
   - **Deliberate Low Bandwidth:** The bandwidth of this loop is intentionally set very low ($10 - 20$ Hz) to prevent the controller from responding to the inevitable 100 Hz or 120 Hz ripple present on V_d.
2. **Analog Multiplier Stage:**
   - Multiplies the slowly varying error signal v_e with a rectified voltage template signal $|v_s(t)|$ obtained directly from a potential divider on the AC mains:

$$i_L^*(t) = K \cdot v_e \cdot |v_s(t)| = \hat{I}_L^* |\sin(\omega t)| \tag{78}$$

   This establishes an instantaneous reference current $i_L^*(t)$ that is guaranteed to have the exact same shape and phase as the AC mains voltage!
3. **Fast Inner Current Loop (Waveform Shaping):**
   - Senses the instantaneous inductor current $i_L(t)$ using a shunt resistor or Hall-effect sensor.
   - A high-bandwidth current controller (Average Current Mode Control or Peak Current Control) forces $i_L(t)$ to track $i_L^*(t)$ cycle-by-cycle at switching frequencies of $50 - 200$ kHz.
   - When switch S is closed: Inductor current rises ($di_L/dt = |v_s|/L$).
   - When switch S is open: Inductor current discharges into the load ($di_L/dt = (|v_s| - V_d)/L < 0$).

$$i_d(t) = I_{\text{load}} + i_c(t) = \frac{V_s I_s}{V_d} - \frac{V_s I_s}{V_d}\cos 2\omega t \tag{18-10}$$

ıere the average value of i_d is

$$I_d = I_{\text{load}} = \frac{V_s I_s}{V_d} \tag{18-11}$$

d the current through the capacitor is

$$i_c(t) = -\frac{V_s I_s}{V_d}\cos 2\omega t = -I_d \cos 2\omega t \tag{18-12}$$

'en though this analysis is carried out by assuming the voltage across the capacitor to be ıple-free dc, the ripple in v_d can be estimated from Eq. 18-12 as

$$v_{d,\text{ripple}}(t) \approx \frac{1}{C_d}\int i_c\, dt = -\frac{I_d}{2\omega C_d}\sin 2\omega t \tag{18-13}$$

ıich can be kept low by selecting a suitably large value of C_d. A series-tuned LC filter ıed for twice the ac frequency according to Eq. 18-12 may be put in parallel with C_d minimize the ripple in the dc voltage. It should be noted that the switching-frequency mponents of currents in i_d and the high-frequency components in the load current will o flow through C_d.

Because the input current to the step-up converter is to be shaped, the step-up nverter is operated in a current-regulated mode, as discussed in Chapter 8 in nnection with dc-to-ac inverters. The feedback control is shown in a block diagram m in Fig. 18-5, where i_L^* is the reference or the desired value of the current i_L in g. 18-4*a*. Here i_L^* has the same waveform as $|v_s|$. The amplitude of i_L^* should be such to maintain the output voltage at a desired or reference level V_d^*, in spite of the riation in load and the fluctuation of the line voltage from its nominal value. The ıveform of i_L^* in Fig. 18-5 is obtained by measuring $|v_s|$ in Fig. 18-4*a* by means of a istive potential divider and multiplying it with the amplified error between the

Figure 13: Control architecture of an Active Power Factor Correction (PFC) boost converter: Outer voltage control loop regulating DC output voltage V_d, analog multiplier generating reference current i_L^*, and inner high-speed current loop driving PWM switch gate. (Source: Ned Mohan, Fig. 18-5)

Key Operating Metrics and Benefits of Active PFC

- **Near-Unity Power Factor:** The input line current $i_s(t)$ is a clean, continuous sinusoid precisely in-phase with the line voltage ($\phi_1 \approx 0 \implies \text{DPF} \approx 1.0$).
- **Ultra-Low Harmonic Distortion:** THD_i is suppressed from $> 80\%$ down to $< 3\% - 5\%$, easily satisfying IEEE 519 limits under all operating tiers!
- **Universal Input Voltage Range:** Active boost PFC automatically accommodates universal AC line inputs ($85\ \text{V}_{\text{rms}} - 265\ \text{V}_{\text{rms}}$, $47 - 63$ Hz) while maintaining a rock-steady regulated 400 V DC bus for downstream inverters or DC-DC stages!

Ten Comprehensive University Exam Solved Numerical Problems

Problem 1: Single-Phase Full-Wave AC Voltage Controller with Resistive Load

Problem 1: ACVC with R Load Design & Analysis

Statement: A single-phase full-wave AC voltage controller is supplied from a 230 V, 50 Hz single-phase source and feeds a purely resistive load of $R = 10\,\Omega$. The firing delay angle is set to $\alpha = \pi/3$ ($60°$). Determine:

1. The RMS output voltage V_{or}.
2. The RMS load current I_{or} and real active power dissipated in the load P_o.
3. The input power factor PF.
4. The average and RMS current ratings required for each individual thyristor.

Solution:

1. **RMS Output Voltage (V_{or}):** Using the master analytical formula for resistive load (equation (6)):

$$V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}$$

Here, $\alpha = \frac{\pi}{3} = 60°$ (1.0472 rad), and $2\alpha = \frac{2\pi}{3} = 120°$.

$$\sin(2\alpha) = \sin(120°) = \frac{\sqrt{3}}{2} \approx 0.8660$$

Substituting the numerical values into the square root expression:

$$\text{Term} = 1 - \frac{\pi/3}{\pi} + \frac{0.8660}{2\pi} = 1 - \frac{1}{3} + \frac{0.8660}{6.2832} = \frac{2}{3} + 0.1378 = 0.6667 + 0.1378 = 0.8045$$

$$\sqrt{0.8045} \approx 0.8969$$

Therefore, the RMS load voltage is:

$$\mathbf{V_{or}} = 230 \times 0.8969 = \mathbf{206.3}\ \text{V}$$

2. **RMS Load Current and Active Load Power:**

$$\mathbf{I_{or}} = \frac{V_{or}}{R} = \frac{206.3\ \text{V}}{10\,\Omega} = \mathbf{20.63}\ \text{A}$$

The real power absorbed by the load resistor is:

$$\mathbf{P_o} = I_{or}^2 \cdot R = (20.63)^2 \times 10 = \mathbf{4256}\ \text{W} = \mathbf{4.256}\ \text{kW}$$

3. **Input Power Factor (PF):**

$$\text{PF} = \frac{P_o}{V_s I_{or}} = \frac{V_{or}}{V_s} = \frac{206.3}{230} = \mathbf{0.897} \text{ (lagging)}$$

4. **Thyristor Current Ratings:** Each thyristor carries current for one half-cycle. Peak supply voltage is $V_m = \sqrt{2} \times 230 = 325.27$ V.

$$\mathbf{I_{T(avg)}} = \frac{V_m}{2\pi R}(1 + \cos\alpha) = \frac{325.27}{2\pi \times 10}(1 + \cos 60^\circ) = \frac{32.527}{6.2832}(1 + 0.5) = 5.177 \times 1.5 = \mathbf{7.77} \text{ A}$$

$$\mathbf{I_{T(rms)}} = \frac{I_{or}}{\sqrt{2}} = \frac{20.63}{\sqrt{2}} = \mathbf{14.59} \text{ A}$$

Problem 2: Single-Phase AC Voltage Controller with Inductive RL Load

Problem 2: ACVC with RL Load and Extinction Angle

Statement: A 230 V, 50 Hz single-phase AC voltage controller feeds a series RL load with $R = 4\,\Omega$ and $\omega L = 3\,\Omega$.

1. Determine the minimum permissible firing angle α_{min} for controllable operation.
2. For a firing angle $\alpha = 60^\circ$, if the numerical extinction angle is found to be $\beta = 220^\circ$ (3.8397 rad), calculate the conduction angle γ and the RMS output voltage V_{or}.

Solution:

1. **Minimum Firing Angle (α_{min}):** The load impedance magnitude and phase angle are:

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = 5.0\,\Omega$$

$$\phi = \arctan\left(\frac{\omega L}{R}\right) = \arctan\left(\frac{3}{4}\right) = 36.87^\circ = 0.6435 \text{ rad}$$

As established in Section 1, to maintain independent phase control and avoid current continuity breakdown, the firing delay angle must satisfy $\alpha \geq \phi$:

$$\alpha_{\text{min}} = \phi = \mathbf{36.87}^\circ$$

2. **Conduction Angle (γ) and RMS Output Voltage (V_{or}):** Given $\alpha = 60^\circ = \frac{\pi}{3}$ rad (1.0472 rad) and $\beta = 220^\circ = 3.8397$ rad:

$$\gamma = \beta - \alpha = 220^\circ - 60^\circ = \mathbf{160}^\circ = \mathbf{2.7925} \text{ rad}$$

Using equation (18):

$$V_{or} = V_s \sqrt{\frac{1}{\pi}\left[(\beta - \alpha) - \frac{\sin(2\beta) - \sin(2\alpha)}{2}\right]}$$

Evaluate the sinusoidal arguments:

$$2\beta = 2 \times 220^\circ = 440^\circ = 80^\circ \implies \sin(440^\circ) = \sin(80^\circ) = 0.9848$$

$$2\alpha = 2 \times 60^\circ = 120^\circ \implies \sin(120^\circ) = 0.8660$$

$$\frac{\sin(2\beta) - \sin(2\alpha)}{2} = \frac{0.9848 - 0.8660}{2} = \frac{0.1188}{2} = 0.0594$$

Now evaluate the term inside the square root:

$$\text{Term} = \frac{1}{\pi}[2.7925 - 0.0594] = \frac{2.7331}{3.1416} = 0.8700$$
$$\sqrt{\text{Term}} = \sqrt{0.8700} \approx 0.9327$$

Therefore, the RMS output voltage is:

$$\mathbf{V_{or}} = 230 \times 0.9327 = \mathbf{214.5}\ \text{V}$$

Problem 3: Single-Phase Full-Bridge Square-Wave Inverter Analysis

Problem 3: Square-Wave Full-Bridge Inverter Analysis

Statement: A single-phase full-bridge inverter is fed from a $48\,\text{V}$ battery bank and delivers power to a series RL load with $R = 5\,\Omega$ and $L = 20\,\text{mH}$ operating at fundamental frequency $f = 50\,\text{Hz}$. Determine:

1. The total RMS output voltage V_{or} and the fundamental RMS output voltage $V_{o1(rms)}$.
2. The Fourier expression for the fundamental load current $i_{o1}(t)$ and its RMS magnitude $I_{o1(rms)}$.
3. The fundamental active power P_1 absorbed by the load.
4. The total harmonic distortion of the output voltage THD_v.

Solution:

1. **Output Voltages:** For a full-bridge square wave inverter:

$$\mathbf{V_{or}} = V_{dc} = \mathbf{48.0}\ \text{V}$$
$$\mathbf{V_{o1(rms)}} = \frac{2\sqrt{2}}{\pi}V_{dc} = \frac{2\sqrt{2}}{\pi} \times 48 = 0.9003 \times 48 = \mathbf{43.21}\ \text{V}$$

   Peak fundamental voltage: $\hat{V}_{o1} = \frac{4V_{dc}}{\pi} = \frac{4\times48}{\pi} = 61.12\,\text{V}$.

2. **Fundamental Load Current:** At $f = 50\,\text{Hz}$, angular frequency $\omega = 2\pi \times 50 = 314.16\,\text{rad/s}$.

$$X_{L1} = \omega L = 314.16 \times 20 \times 10^{-3} = 6.283\,\Omega$$
$$Z_1 = \sqrt{R^2 + X_{L1}^2} = \sqrt{5^2 + (6.283)^2} = \sqrt{25 + 39.48} = \sqrt{64.48} = 8.03\,\Omega$$
$$\phi_1 = \arctan\left(\frac{6.283}{5}\right) = \arctan(1.2566) = 51.49^\circ = 0.8987\,\text{rad}$$

   Peak fundamental load current:

$$\hat{I}_{o1} = \frac{\hat{V}_{o1}}{Z_1} = \frac{61.12\,\text{V}}{8.03\,\Omega} = 7.61\,\text{A} \implies \mathbf{I_{o1(rms)}} = \frac{7.61}{\sqrt{2}} = \mathbf{5.38}\,\text{A}$$

   The instantaneous fundamental current expression is:

$$\mathbf{i_{o1}(t)} = \mathbf{7.61}\sin(\mathbf{314.16t} - \mathbf{51.49}^\circ)\ \text{A}$$

3. **Fundamental Active Power (P_1):**

$$\mathbf{P_1} = I_{o1(rms)}^2 \cdot R = (5.38)^2 \times 5 = 28.94 \times 5 = \mathbf{144.7}\ \text{W}$$

4. **Total Harmonic Distortion (THD$_v$):**

$$\text{THD}_\mathbf{v} = \sqrt{\frac{V_{or}^2 - V_{o1(rms)}^2}{V_{o1(rms)}^2}} = \sqrt{\left(\frac{48}{43.21}\right)^2 - 1} = \sqrt{1.2337 - 1} = \mathbf{48.34\%}$$

Problem 4: SPWM Full-Bridge Inverter with Bipolar Voltage Switching

Problem 4: Bipolar SPWM Inverter Parameter Sizing

Statement: A single-phase full-bridge inverter utilizing sinusoidal PWM with bipolar switching is supplied from a $V_{dc} = 300\,\text{V}$ DC rail. The carrier frequency is $f_s = 2.0\,\text{kHz}$ and the desired fundamental output frequency is $f_o = 50\,\text{Hz}$. The amplitude of the triangular carrier is $\hat{V}_{\text{tri}} = 5\,\text{V}$, and the amplitude of the control reference is $\hat{V}_{\text{control}} = 4\,\text{V}$.

1. Determine the amplitude modulation index m_a and frequency modulation ratio m_f.
2. Calculate the peak and RMS fundamental output voltages.
3. Identify the frequencies of the first four dominant harmonic components in the output voltage.

Solution:

1. **Modulation Indices:**

$$\mathbf{m_a} = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}} = \frac{4\,\text{V}}{5\,\text{V}} = \mathbf{0.80}$$

$$\mathbf{m_f} = \frac{f_s}{f_o} = \frac{2000\,\text{Hz}}{50\,\text{Hz}} = \mathbf{40}$$

   Design Note: In practical design, m_f would be selected as an odd integer (e.g., 39 or 41) to ensure half-wave symmetry and eliminate even harmonics.

2. **Fundamental Output Voltages:** Since $m_a = 0.80 \leq 1.0$, the inverter operates in the linear modulation range:

$$\hat{\mathbf{V}}_{\mathbf{o1}} = m_a \cdot V_{dc} = 0.80 \times 300\,\text{V} = \mathbf{240.0}\,\text{V}$$

$$\mathbf{V_{o1(rms)}} = \frac{\hat{V}_{o1}}{\sqrt{2}} = \frac{240}{\sqrt{2}} = \mathbf{169.7}\,\text{V}$$

3. **Dominant Harmonic Frequencies:** For bipolar switching, harmonics appear as sidebands centered around m_f and its multiples:
   - Center harmonic: $f_{m_f} = m_f \cdot f_o = 40 \times 50 = \mathbf{2000}$ Hz.
   - Sidebands around m_f: $(m_f \pm 2) f_o = (40 \pm 2) \times 50 \implies \mathbf{1900}$ Hz and $\mathbf{2100}$ Hz.
   - Sidebands around $2m_f$: $2m_f \cdot f_o = 4000$ Hz; sidebands at $(2m_f \pm 1) f_o \implies \mathbf{3950}$ Hz and $\mathbf{4050}$ Hz.

Problem 5: SPWM Full-Bridge Inverter with Unipolar Voltage Switching

Problem 5: Unipolar Switching vs. Bipolar Output Filter Sizing

Statement: The inverter of Problem 4 is reconfigured to operate with **unipolar voltage switching** with the same DC link voltage ($V_{dc} = 300$ V), $m_a = 0.80$, and carrier frequency $f_s = 2$ kHz.

1. Determine the peak and RMS fundamental output voltages.
2. What is the lowest harmonic frequency present in the unipolar output voltage?
3. If an LC low-pass filter with corner frequency $f_c = \frac{1}{2\pi\sqrt{LC}}$ is designed to be one decade below the lowest harmonic frequency, calculate the required LC product for both bipolar and unipolar PWM.

Solution:

1. **Fundamental Voltages:** The fundamental output voltage in unipolar switching is identical to bipolar switching:

$$\hat{\mathbf{V}}_{\mathbf{o1}} = m_a \cdot V_{dc} = 0.80 \times 300 = \mathbf{240}\ \text{V}$$
$$\mathbf{V_{o1(rms)}} = \frac{240}{\sqrt{2}} = \mathbf{169.7}\ \text{V}$$

2. **Lowest Harmonic Frequency:** In unipolar switching, all harmonics centered around m_f (2000 Hz) cancel out! The first cluster appears around $2m_f$:

$$f_{\text{lowest}} = (2m_f - 1)f_o = (80 - 1) \times 50 = 79 \times 50 = \mathbf{3950}\ \text{Hz} \approx \mathbf{4.0}\ \text{kHz}$$

3. LC **Filter Product Comparison:**

   - **Bipolar PWM:** Lowest harmonic occurs around $f_s = 2000$ Hz. One decade below gives filter cutoff $f_{c,\text{bipolar}} = 200$ Hz:

$$(LC)_{\text{bipolar}} = \frac{1}{(2\pi f_{c,\text{bipolar}})^2} = \frac{1}{(2\pi \times 200)^2} = \frac{1}{(1256.6)^2} = \mathbf{6.33 \times 10^{-7}}\ \text{s}^2$$

   - **Unipolar PWM:** Lowest harmonic occurs around $2f_s = 4000$ Hz. One decade below gives filter cutoff $f_{c,\text{unipolar}} = 400$ Hz:

$$(LC)_{\text{unipolar}} = \frac{1}{(2\pi \times 400)^2} = \frac{1}{(2513.3)^2} = \mathbf{1.58 \times 10^{-7}}\ \text{s}^2$$

   Conclusion: Unipolar switching allows a **4× reduction in the LC product**, dramatically reducing filter inductor weight, copper losses, and capacitor cost!

Problem 6: Three-Phase PWM Voltage Source Inverter

Problem 6: Three-Phase SPWM Line-to-Line Voltage Calculation

Statement: A three-phase bridge inverter is supplied from an electric vehicle battery pack of $V_{dc} = 400$ V. The inverter is controlled using three-phase sinusoidal PWM in the linear modulation range with $m_a = 0.85$ and $m_f = 21$.

1. Calculate the peak and RMS values of the fundamental phase-to-neutral (pole) voltage.
2. Calculate the peak and RMS values of the fundamental line-to-line output voltage.
3. Calculate the maximum achievable RMS fundamental line-to-line voltage in the linear range ($m_a = 1.0$), and compare it to the six-step square-wave mode.

Solution:

1. **Fundamental Phase Voltage (v_{Ao1}):**

$$\hat{\mathbf{V}}_{\mathbf{Ao1}} = m_a \frac{V_{dc}}{2} = 0.85 \times \frac{400}{2} = \mathbf{170.0}\ \text{V}$$

$$\mathbf{V}_{\mathbf{Ao1(rms)}} = \frac{\hat{V}_{Ao1}}{\sqrt{2}} = \frac{170}{\sqrt{2}} = \mathbf{120.2}\ \text{V}$$

2. **Fundamental Line-to-Line Voltage (V_{LL1}):** Using equations (51) and (52):

$$\hat{\mathbf{V}}_{\mathbf{LL1}} = \sqrt{3}\hat{V}_{Ao1} = \sqrt{3} \times 170.0\ \text{V} = \mathbf{294.4}\ \text{V}$$

$$\mathbf{V}_{\mathbf{LL1(rms)}} = \frac{\sqrt{6}}{4} m_a V_{dc} = 0.6124 \times 0.85 \times 400 = \mathbf{208.2}\ \text{V}$$

3. **Maximum Voltage Comparison:**

- **Maximum Linear SPWM ($m_a = 1.0$):**

$$V_{LL1(rms),\text{max}} = 0.6124 \times 400\ \text{V} = \mathbf{245.0}\ \text{V}$$

- **Six-Step Square-Wave Mode:**

$$V_{LL1(rms),\text{square}} = \frac{\sqrt{6}}{\pi} V_{dc} = \frac{\sqrt{6}}{\pi} \times 400 = 0.7797 \times 400 = \mathbf{311.9}\ \text{V}$$

Operating in six-step mode yields $\frac{311.9}{245.0} = 1.273$ (+27.3%) higher fundamental voltage than linear SPWM.

Problem 7: Third Harmonic Injection PWM (THIPWM) Voltage Boost

Problem 7: THIPWM DC Bus Utilization Advantage

Statement: A 3-phase inverter fed by a $V_{dc} = 600$ V DC bus supplies a 415 V (RMS line-to-line), 50 Hz three-phase induction motor.

1. Determine if standard sinusoidal PWM in the linear range can deliver the required 415 V RMS line-to-line motor voltage.
2. Calculate the maximum line-to-line RMS voltage achievable using Third Harmonic Injection PWM (THIPWM) while remaining strictly within the linear modulation boundary.
3. Can THIPWM supply the motor at rated voltage without overmodulation?

Solution:

1. **Maximum Linear Voltage with Standard SPWM:** For standard SPWM, the maximum modulation index is $m_a = 1.0$:

$$V_{LL1(rms),\max\,(\text{SPWM})} = 0.6124 \times V_{dc} = 0.6124 \times 600\ \text{V} = \mathbf{367.4}\ \text{V}$$

Since $367.4\ \text{V} < 415\ \text{V}$, **standard SPWM cannot provide the rated voltage** without entering severe overmodulation distortion!

2. **Maximum Linear Voltage with THIPWM:** With optimal third harmonic injection ($v_3 = \frac{1}{6}\hat{V}_1 \sin 3\omega t$), the maximum fundamental voltage in the linear range is boosted by $\frac{2}{\sqrt{3}} \approx 1.1547$:

$$\mathbf{V_{LL1(rms),\max\,(\text{THIPWM})}} = \frac{V_{dc}}{\sqrt{2}} = \frac{600\ \text{V}}{\sqrt{2}} = \mathbf{424.26}\ \text{V}$$

3. **Feasibility Assessment:** Since $424.26\ \text{V} > 415\ \text{V}$, **THIPWM comfortably delivers the rated 415 V output** with $m_a = \frac{415}{424.26} = 0.978 \leq 1.0$, completely avoiding low-order harmonic distortion!

Problem 8: Analytical Evaluation of Dead-Time (Blanking Time) Voltage Drop

Problem 8: Dead-Time Induced Fundamental Voltage Reduction

Statement: A high-power IGBT three-phase inverter operates from a $V_{dc} = 750$ V DC bus with switching frequency $f_s = 10$ kHz. To prevent shoot-through, a blanking time of $t_d = 2.5\,\mu$s is programmed into the gate drive controller.

1. Calculate the average voltage error ΔV_o per phase leg.
2. Determine the percentage reduction in fundamental output voltage if the inverter is operating at modulation index $m_a = 0.20$ (low-speed motor operation).
3. Propose how this distortion can be compensated in industrial DSP motor drives.

Solution:

1. **Average Leg Voltage Error (ΔV_o):** Using equation (44):

$$\Delta\mathbf{V_o} = 4 f_s t_d V_{dc} = 4 \times (10 \times 10^3\ \text{Hz}) \times (2.5 \times 10^{-6}\ \text{s}) \times 750\ \text{V}$$

$$\Delta\mathbf{V_o} = 4 \times 0.025 \times 750 = 0.10 \times 750 = \mathbf{75.0}\ \text{V}$$

2. **Percentage Voltage Reduction at $m_a = 0.20$:** Nominal fundamental peak pole voltage without dead time:

$$\hat{V}_{Ao1} = m_a \frac{V_{dc}}{2} = 0.20 \times \frac{750}{2} = 75.0\ \text{V}$$

The fundamental component of the dead-time error square wave is $\frac{4}{\pi}\Delta V_o$:

$$\hat{V}_{\text{error},1} = \frac{4}{\pi}(f_s t_d V_{dc}) = \frac{4}{\pi} \times 18.75 = 23.87\ \text{V}$$

Percentage fundamental loss:

$$\%\ \text{Loss} = \frac{23.87\ \text{V}}{75.0\ \text{V}} \times 100\% = \mathbf{31.8}\%!$$

At low speeds ($m_a = 0.20$), dead-time causes an enormous 31.8% drop in motor flux and severe 5th/7th torque pulsations!

3. **Compensation Strategy:** DSP controllers implement **Dead-Time Compensation**: the controller detects the sign of the phase current ($i_A > 0$ or $i_A < 0$) using current sensors and dynamically adds or subtracts a pulse width correction $\Delta t = t_d$ to the PWM timer registers, completely restoring fundamental voltage linearity!

Problem 9: Power Factor and THD in Non-Sinusoidal Rectifiers

Problem 9: True Power Factor and Distortion Factor Evaluation

Statement: A single-phase diode bridge rectifier with a capacitive filter draws a non-sinusoidal input current from a 230 V, 50 Hz supply. A power quality analyzer records the fundamental current component as $I_{s1} = 10$ A (RMS) lagging the line voltage by $\phi_1 = 18°$, and harmonic current components of: $I_{s3} = 7.5$ A, $I_{s5} = 4.5$ A, $I_{s7} = 2.0$ A, and $I_{s9} = 1.0$ A (higher harmonics negligible).

1. Calculate the total RMS input current I_s.
2. Compute the current Total Harmonic Distortion (THD$_i$).
3. Determine the Distortion Factor (K_{dist}) and Displacement Power Factor (DPF).
4. Calculate the True Power Factor (PF) and real active power drawn from the mains.

Solution:

1. **Total RMS Current (I_s):**

$$I_{\text{dist}}^2 = \sum_{h=3,5,7,9} I_{sh}^2 = (7.5)^2 + (4.5)^2 + (2.0)^2 + (1.0)^2 = 56.25 + 20.25 + 4.0 + 1.0 = 81.5 \text{ A}^2$$

$$I_s = \sqrt{I_{s1}^2 + I_{\text{dist}}^2} = \sqrt{10^2 + 81.5} = \sqrt{181.5} = \mathbf{13.47} \text{ A}$$

2. **Current Total Harmonic Distortion (THD$_i$):**

$$\text{THD}_\mathbf{i} = \frac{\sqrt{I_{\text{dist}}^2}}{I_{s1}} = \frac{\sqrt{81.5}}{10} = \frac{9.028}{10} = 0.9028 \implies \mathbf{90.28\%}$$

3. **Distortion Factor and Displacement Factor:**

$$\mathbf{K}_{\text{dist}} = \frac{I_{s1}}{I_s} = \frac{10 \text{ A}}{13.47 \text{ A}} = \mathbf{0.7424}$$

$$\text{DPF} = \cos\phi_1 = \cos(18°) = \mathbf{0.9511}$$

4. **True Power Factor and Real Power:** Using equation (71):

$$\text{PF} = K_{\text{dist}} \times \text{DPF} = 0.7424 \times 0.9511 = \mathbf{0.706} \text{ (lagging)}$$

Verification using equation (74):

$$\text{PF} = \frac{\cos(18°)}{\sqrt{1 + (0.9028)^2}} = \frac{0.9511}{\sqrt{1 + 0.815}} = \frac{0.9511}{\sqrt{1.815}} = \frac{0.9511}{1.3472} = 0.706 \text{ (Exact match!)}$$

Real active power consumed:

$$\mathbf{P} = V_s I_{s1} \cos\phi_1 = 230 \times 10 \times 0.9511 = \mathbf{2187.5}\ \text{W} = \mathbf{2.188}\ \text{kW}$$

Apparent power: $S = V_s I_s = 230 \times 13.47 = 3098$ VA. PF $= 2187.5/3098 = 0.706$.

Problem 10: Active Power Factor Correction (PFC) Boost Converter Design

Problem 10: Active PFC Boost Inductor Sizing & IEEE 519 Check

Statement: A 1.5 kW single-phase Active PFC boost converter is connected to a $230\ V_{rms}$, 50 Hz AC line. The regulated DC bus voltage is $V_d = 400$ V, switching frequency is $f_s = 100$ kHz, and target converter efficiency is $\eta = 95\%$.

1. Determine the peak and RMS fundamental input currents drawn from the AC line.
2. Calculate the maximum peak-to-peak inductor ripple current $\Delta I_{L,\max}$ if ripple is limited to 20% of peak input current.
3. Calculate the required boost inductance L.
4. At the customer Point of Common Coupling (PCC), the available short-circuit current is $I_{sc} = 650$ A. What is the maximum permissible total demand distortion (TDD) specified by IEEE 519?

Solution:

1. **Input Line Currents:** Input active power drawn from the mains:

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{1500\ \text{W}}{0.95} = 1578.9\ \text{W}$$

Since the PFC converter shapes current to achieve near-unity power factor (PF ≈ 1.0):

$$\mathbf{I_{s(rms)}} = \frac{P_{\text{in}}}{V_s} = \frac{1578.9\ \text{W}}{230\ \text{V}} = \mathbf{6.86}\ \text{A}$$
$$\hat{\mathbf{I}}_{\mathbf{s}} = \sqrt{2} \times I_{s(rms)} = \sqrt{2} \times 6.86 = \mathbf{9.71}\ \text{A}$$

2. **Maximum Inductor Ripple Current ($\Delta I_{L,\max}$):**

$$\Delta\mathbf{I_{L,max}} = 20\% \times \hat{I}_s = 0.20 \times 9.71\ \text{A} = \mathbf{1.942}\ \text{A}$$

3. **Boost Inductance Sizing (L):** In a boost PFC converter, the duty cycle varies continuously across the line cycle: $d(t) = 1 - \frac{|v_s(t)|}{V_d}$. The ripple current is:

$$\Delta i_L = \frac{|v_s| \cdot d}{f_s L} = \frac{|v_s|}{f_s L}\left(1 - \frac{|v_s|}{V_d}\right)$$

Differentiating with respect to $|v_s|$ shows that maximum ripple occurs precisely when $|v_s| = \frac{V_d}{2} = \frac{400}{2} = 200$ V (where duty cycle $d = 0.5$):

$$\Delta I_{L,\max} = \frac{V_d/2 \cdot 0.5}{f_s L} = \frac{V_d}{4 f_s L}$$

Rearranging to solve for inductance L:

$$\mathbf{L} = \frac{V_d}{4 f_s \Delta I_{L,\max}} = \frac{400\ \text{V}}{4 \times (100 \times 10^3\ \text{Hz}) \times 1.942\ \text{A}}$$
$$= \frac{400}{776,800} = 5.15 \times 10^{-4}\ \text{H} = \mathbf{0.515}\ \text{mH} \approx \mathbf{520}\ \mu\text{H}$$

4. **IEEE 519 TDD Limit Evaluation:** The maximum fundamental load current is $I_L = I_{s(rms)} = 6.86$ A. The Short-Circuit Ratio (SCR) at the PCC is:

$$\text{SCR} = \frac{I_{sc}}{I_L} = \frac{650\ \text{A}}{6.86\ \text{A}} = \mathbf{94.75}$$

From Table 3, for $50 < \frac{I_{sc}}{I_L} < 100$:

$$\text{Maximum Permissible TDD} = \mathbf{12.0}\%$$

Since active PFC achieves $\text{THD}_i < 5\%$, the installation effortlessly complies with IEEE 519 with massive safety margins!

Master Formula Sheet & Quick Revision Tables

Master Analytical Formula Matrix

Essential Examination Quick-Tips for KTU S5 EEE Students

1. **When asked for ACVC RMS Voltage Derivation:** Always state the half-wave symmetry condition ($v_o(\omega t + \pi) = -v_o(\omega t)$), integrate $[V_m \sin \omega t]^2$ from α to π (or α to β for RL load), convert $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, and show every intermediate integration step clearly to secure full marks.

2. **Why $\alpha < \phi$ is prohibited in RL ACVC:** Remember the physical mechanism: If $\alpha < \phi$, the current in the previous thyristor is still flowing when the next thyristor is triggered. Conduction becomes continuous, phase control is completely lost, and the converter acts as a short circuit directly connecting the load to the AC mains.

3. **Fourier Series of Square Wave Inverter:** Write down the general half-wave symmetry property first. State why all even harmonics and DC term are zero ($a_0 = 0, a_n = 0, b_{\text{even}} = 0$). Write the general formula $\sum \frac{4V_{dc}}{n\pi} \sin(n\omega t)$ and calculate $\text{THD}_v = \sqrt{\frac{\pi^2}{8} - 1} = 48.34\%$.

4. **Bipolar vs. Unipolar PWM Comparison:** Draw the two-level vs. three-level output waveforms side-by-side. Highlight the three killer facts: (1) Effective switching frequency doubles to $2f_s$; (2) Harmonics around m_f cancel out completely; (3) Output filter size drops by $4\times$.

5. **Why Triplen Harmonics Disappear in 3-Phase Inverters:** State that triplen harmonics ($3rd, 9th, 15th, \ldots$) form a zero-sequence set—they are exactly equal in magnitude and in phase in all three inverter legs ($v_{Ao,3} = v_{Bo,3} = v_{Co,3}$). When computing line-to-line voltage $v_{AB} = v_{Ao} - v_{Bo}$, they subtract to exactly zero!

6. **Dead-Time Derivation:** Draw the leg diagram showing S_1, S_4 and diodes D_1, D_4. Explain that when both switches are off during t_d, the load current i_o forces either D_1 or D_4 to conduct depending solely on the sign of i_o, subtracting or adding volt-seconds to the output pulse.

Table 5: Module 3 Master Analytical Formula Cheat-Sheet for Fast Exam Revision

Topic / Circuit	Key Formulas / Governing Equations	Exam Tips & Critical Insights
ACVC (R Load)	$V_{or} = V_s\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$ $\text{PF} = \frac{V_{or}}{V_s} = \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}}$	At $\alpha = 0 \implies V_{or} = V_s, \text{PF} = 1.0$. At $\alpha = 90° \implies V_{or} = 0.707V_s, \text{PF} = 0.707$. At $\alpha = 180° \implies V_{or} = 0, \text{PF} = 0$.
ACVC (RL Load)	$i_o(\omega t) = \frac{V_m}{Z}[\sin(\omega t - \phi) - \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\omega t - \alpha)}]$ $\sin(\beta - \phi) = \sin(\alpha - \phi)e^{-\frac{R}{\omega L}(\beta - \alpha)}$ $V_{or} = V_s\sqrt{\frac{1}{\pi}[(\beta - \alpha) - \frac{\sin 2\beta - \sin 2\alpha}{2}]}$	**Rule:** $\alpha \geq \phi = \arctan(\omega L/R)$. If $\alpha = \phi \implies \beta = \pi + \alpha \implies \gamma = 180°$ (continuous sinusoidal current). Requires high-frequency pulse train gating!
ACVC (Integral Cycle)	$k = \frac{n}{n+m}, \quad V_{or} = V_s\sqrt{k}$ $\text{PF} = \sqrt{k}$	Zero switching EMI ($dv/dt = 0, di/dt = 0$). Causes line flicker/sub-harmonics. Used for thermal loads.
Half-Bridge VSI (Square Wave)	$v_o = \pm V_{dc}/2, \quad V_{or} = V_{dc}/2$ $\hat{V}_{o1} = \frac{2}{\pi}V_{dc}, \quad V_{o1(rms)} = \frac{\sqrt{2}}{\pi}V_{dc} \approx 0.45V_{dc}$	Requires 3-wire DC split supply. Switch voltage rating $= V_{dc}$ ($2\times$ peak output).
Full-Bridge VSI (Square Wave)	$v_o = \pm V_{dc}, \quad V_{or} = V_{dc}$ $\hat{V}_{o1} = \frac{4}{\pi}V_{dc}, \quad V_{o1(rms)} = \frac{2\sqrt{2}}{\pi}V_{dc} \approx 0.90V_{dc}$ $\text{THD}_v = \sqrt{\frac{\pi^2}{8} - 1} = 48.34\%$	Odd harmonics only ($n = 1, 3, 5, 7, \ldots$). Harmonic amplitude $\hat{V}_{on} = \hat{V}_{o1}/n$. Twice the voltage, $4\times$ power of half-bridge!
Single-Phase Bipolar SPWM	$m_a = \frac{\hat{V}_{\text{control}}}{\hat{V}_{\text{tri}}}, \quad m_f = \frac{f_s}{f_o}$ $\hat{V}_{o1} = m_a V_{dc}, \quad V_{o1(rms)} = \frac{m_a V_{dc}}{\sqrt{2}}$	m_f must be an odd integer (eliminates even harmonics). Lowest harmonics clustered at $m_f \pm 2$.
Single-Phase Unipolar SPWM	$\hat{V}_{o1} = m_a V_{dc}, \quad V_{o1(rms)} = \frac{m_a V_{dc}}{\sqrt{2}}$ Effective switching frequency $= 2f_s$	3-level output ($+V_{dc}, 0, -V_{dc}$). Harmonics around m_f cancel! First cluster at $2m_f$. LC filter product reduced by $4\times$!
Three-Phase SPWM (Linear)	$\hat{V}_{Ao1} = m_a\frac{V_{dc}}{2}$ $\hat{V}_{LL1} = \frac{\sqrt{3}}{2}m_a V_{dc} \approx 0.866 m_a V_{dc}$ $V_{LL1(rms)} = \frac{\sqrt{6}}{4}m_a V_{dc} \approx 0.6124 m_a V_{dc}$	Triplen harmonics ($3rd, 9th, 15th, \ldots$) cancel in line-line! Choose m_f as an odd multiple of 3 ($9, 15, 21, \ldots$). Max linear RMS line voltage $= 0.6124V_{dc}$.
Three-Phase Six-Step (Square)	$V_{LL(rms)} = \sqrt{\frac{2}{3}}V_{dc} \approx 0.8165V_{dc}$ $V_{LL1(rms)} = \frac{\sqrt{6}}{\pi}V_{dc} \approx 0.7797V_{dc}$	$180°$ conduction. Output contains $n = 6k \pm 1$ ($5th, 7th, 11th$). Fundamental is 27.3% higher than max linear SPWM.
Third Harmonic Injection (THIPWM)	$v_{\text{control}} = \hat{V}_1[\sin(\omega t) + \frac{1}{6}\sin(3\omega t)]$ $V_{LL1(rms),\text{max}} = \frac{V_{dc}}{\sqrt{2}} \approx 0.707V_{dc}$	Flattens sine peak by 0.866. **Boosts linear output voltage by 15.5%!** Third harmonic cancels out

7. **IEEE 519 PCC Rule:** Remember that IEEE 519 applies strictly at the **Point of Common Coupling (PCC)**, NOT inside the customer plant. Limits are specified in terms of **Total Demand Distortion (TDD)** and depend directly on the short-circuit ratio I_{sc}/I_L.